

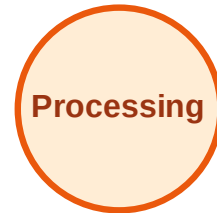
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing

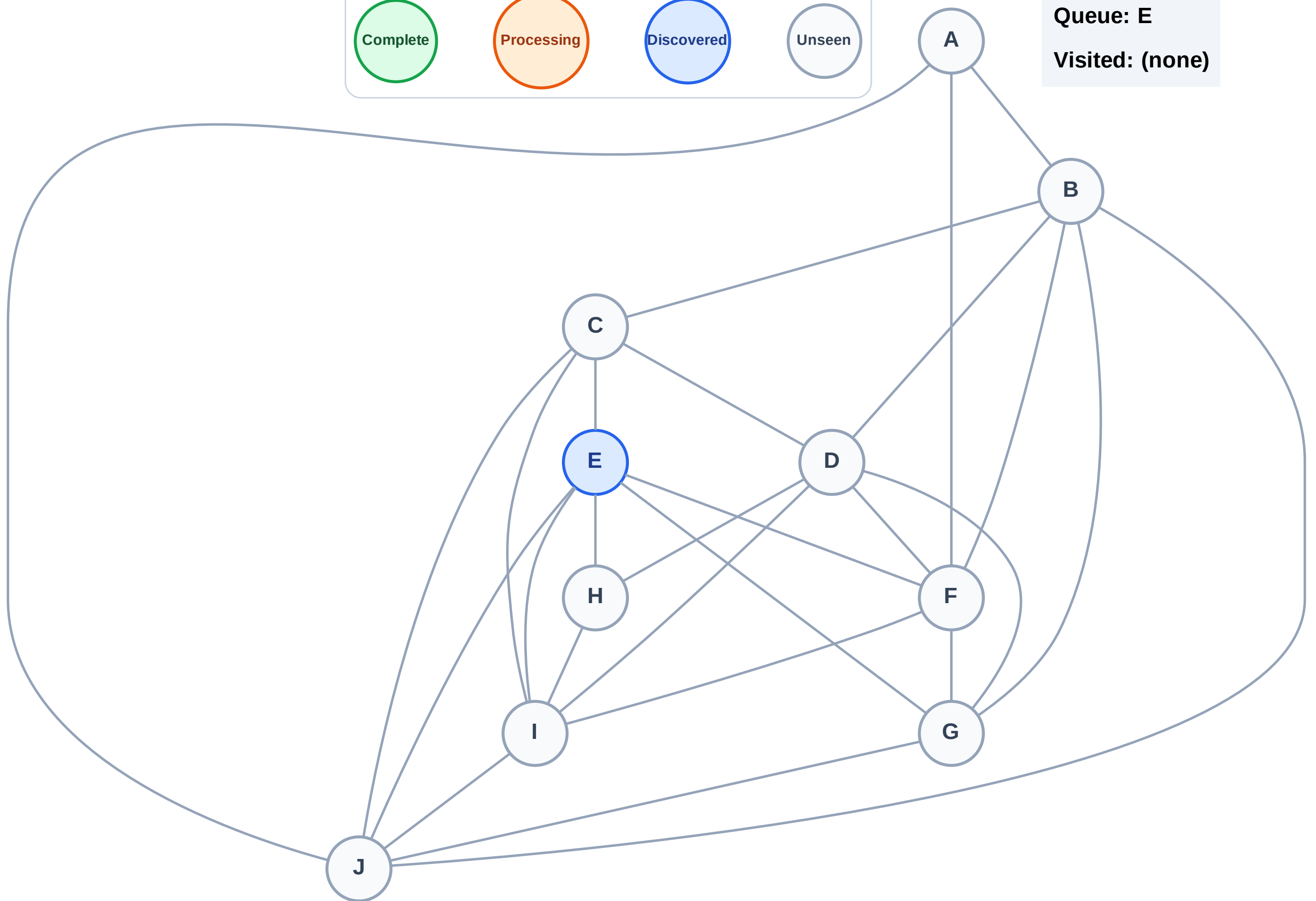


Discovered



Unseen

Queue: E
Visited: (none)



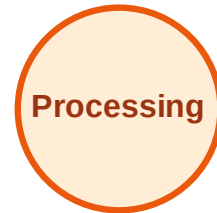
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



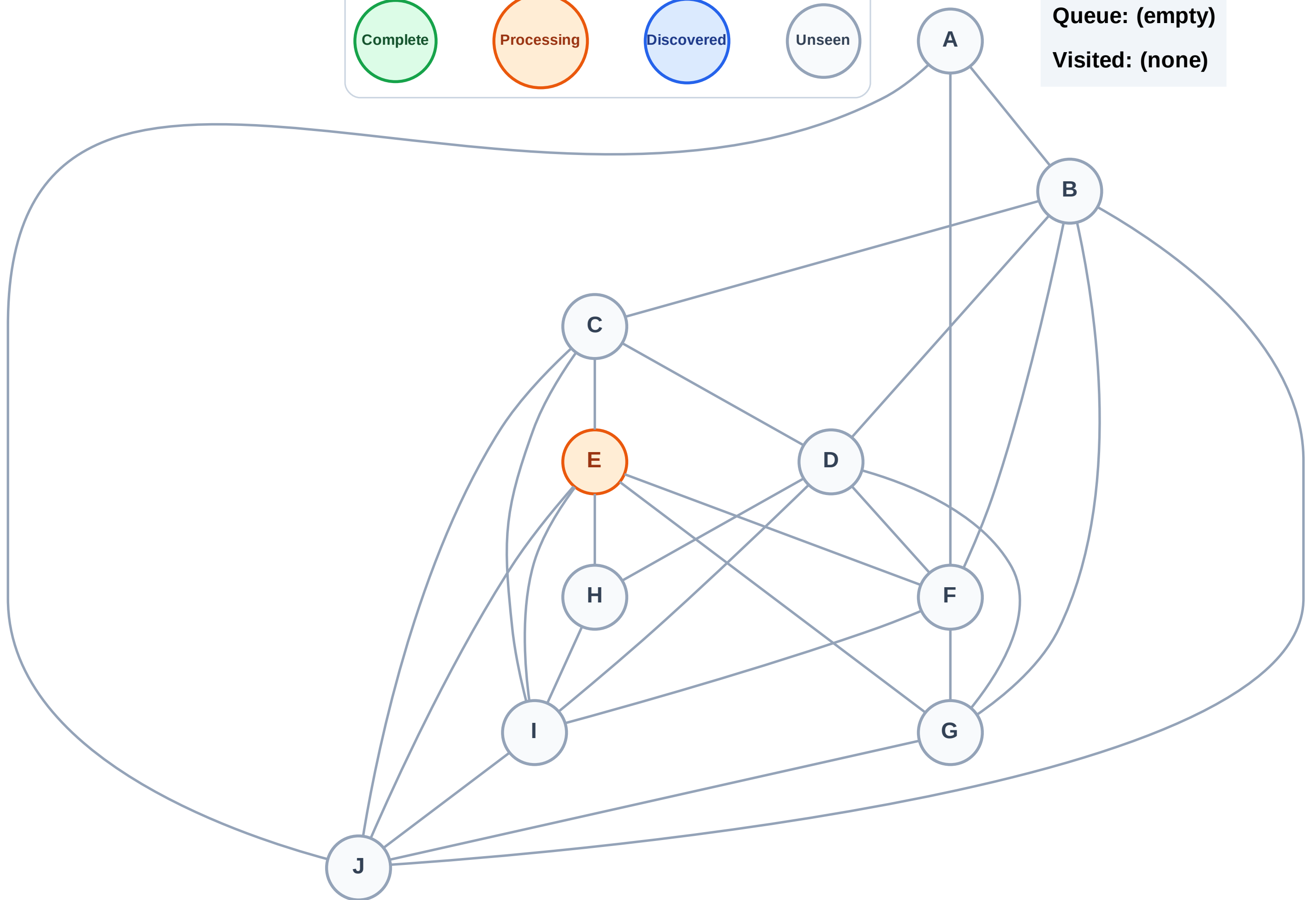
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover C, F, G, H, I, J from E

Legend

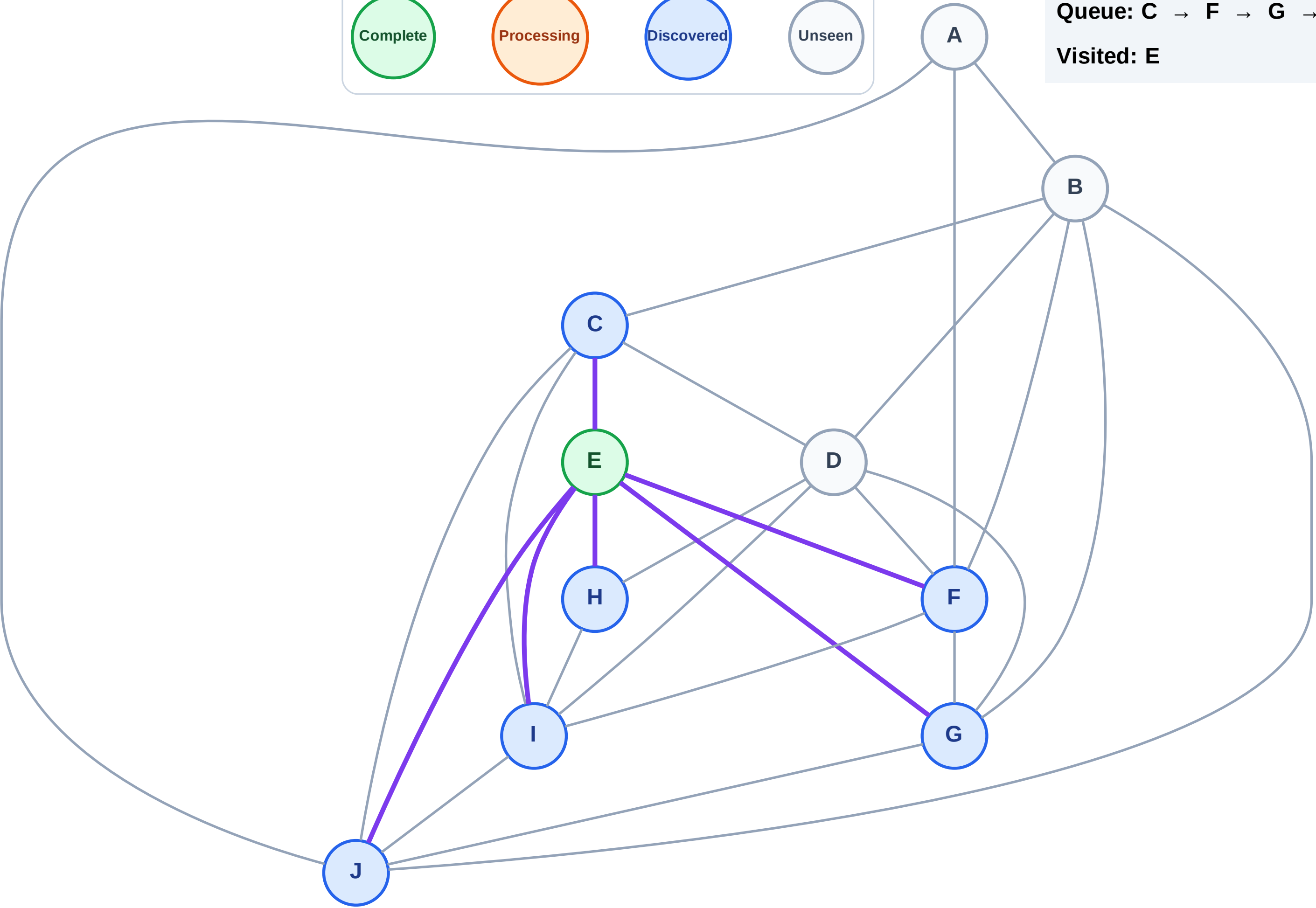
Complete

Processing

Discovered

Unseen

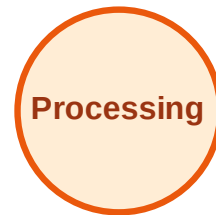
Queue: C → F → G → H → I → J
Visited: E



BFS Traversal

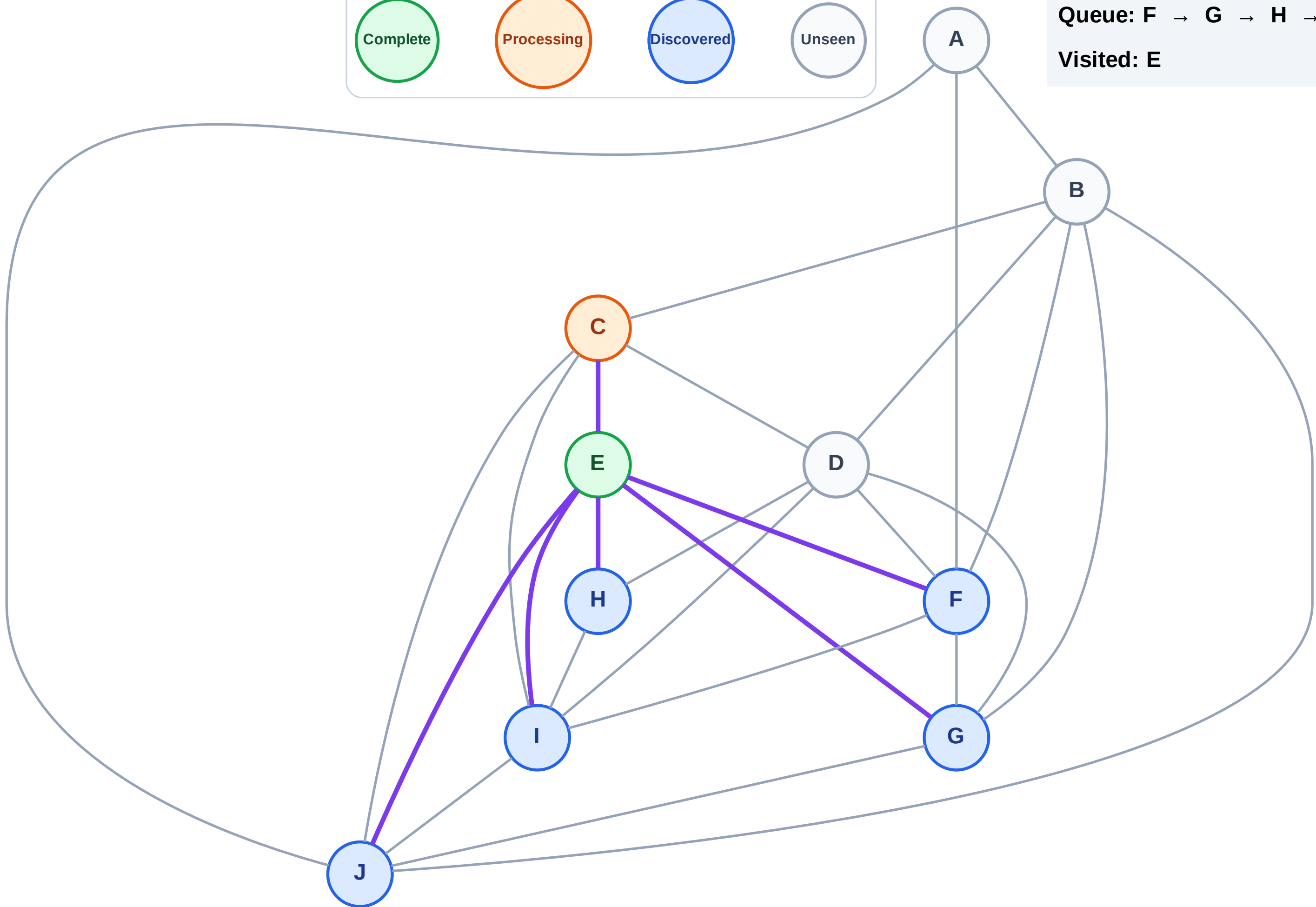
Step 4: Process node C

Legend



Queue: F → G → H → I → J

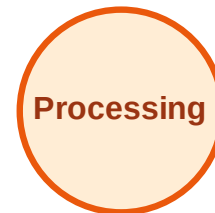
Visited: E



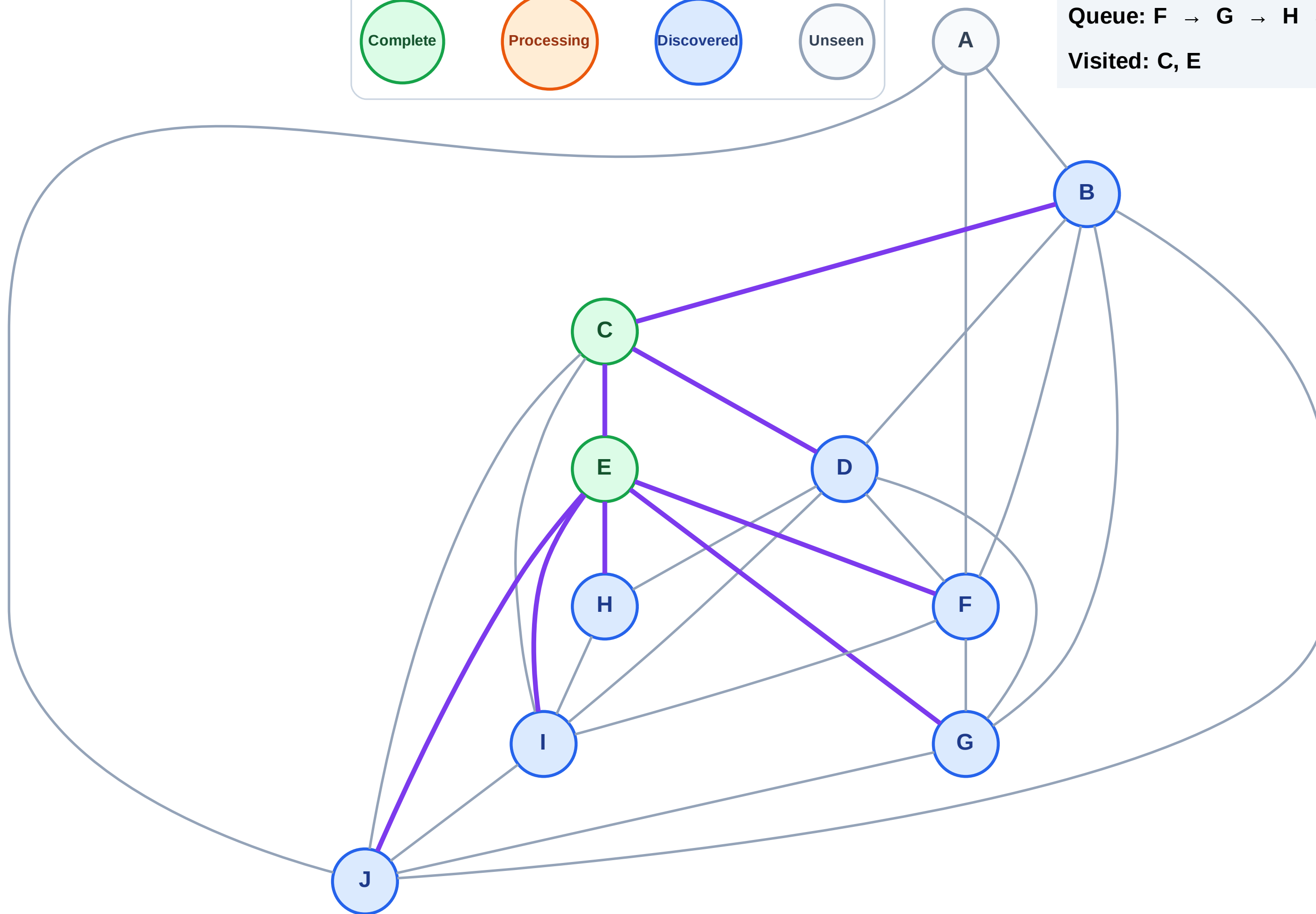
BFS Traversal

Step 5: Discover B, D from C

Legend



Queue: F → G → H → I → J → B → D
Visited: C, E



BFS Traversal

Step 6: Process node F

Legend

Complete

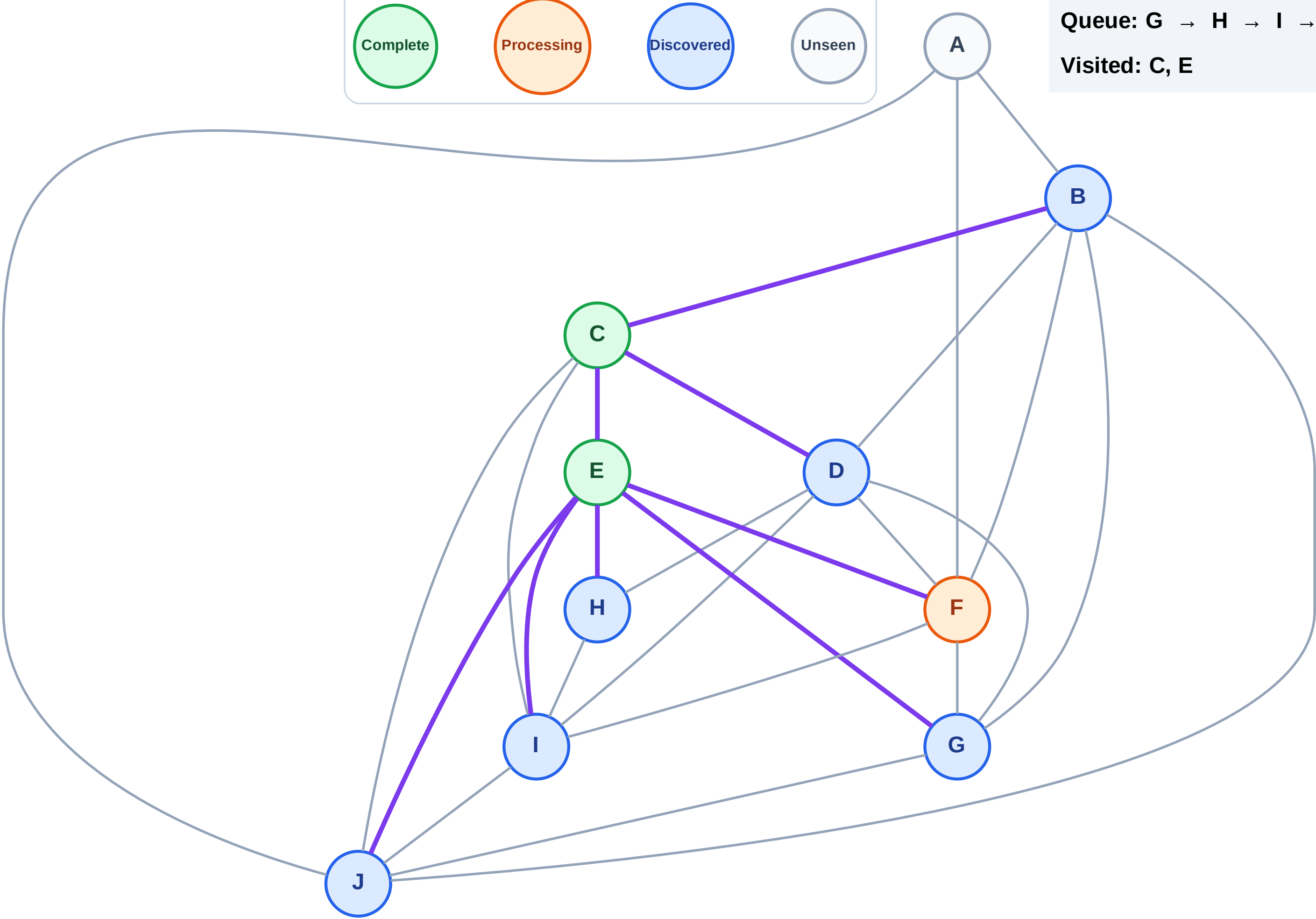
Processing

Discovered

Unseen

Queue: G → H → I → J → B → D

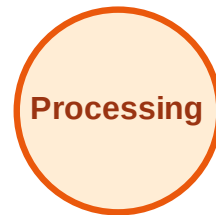
Visited: C, E



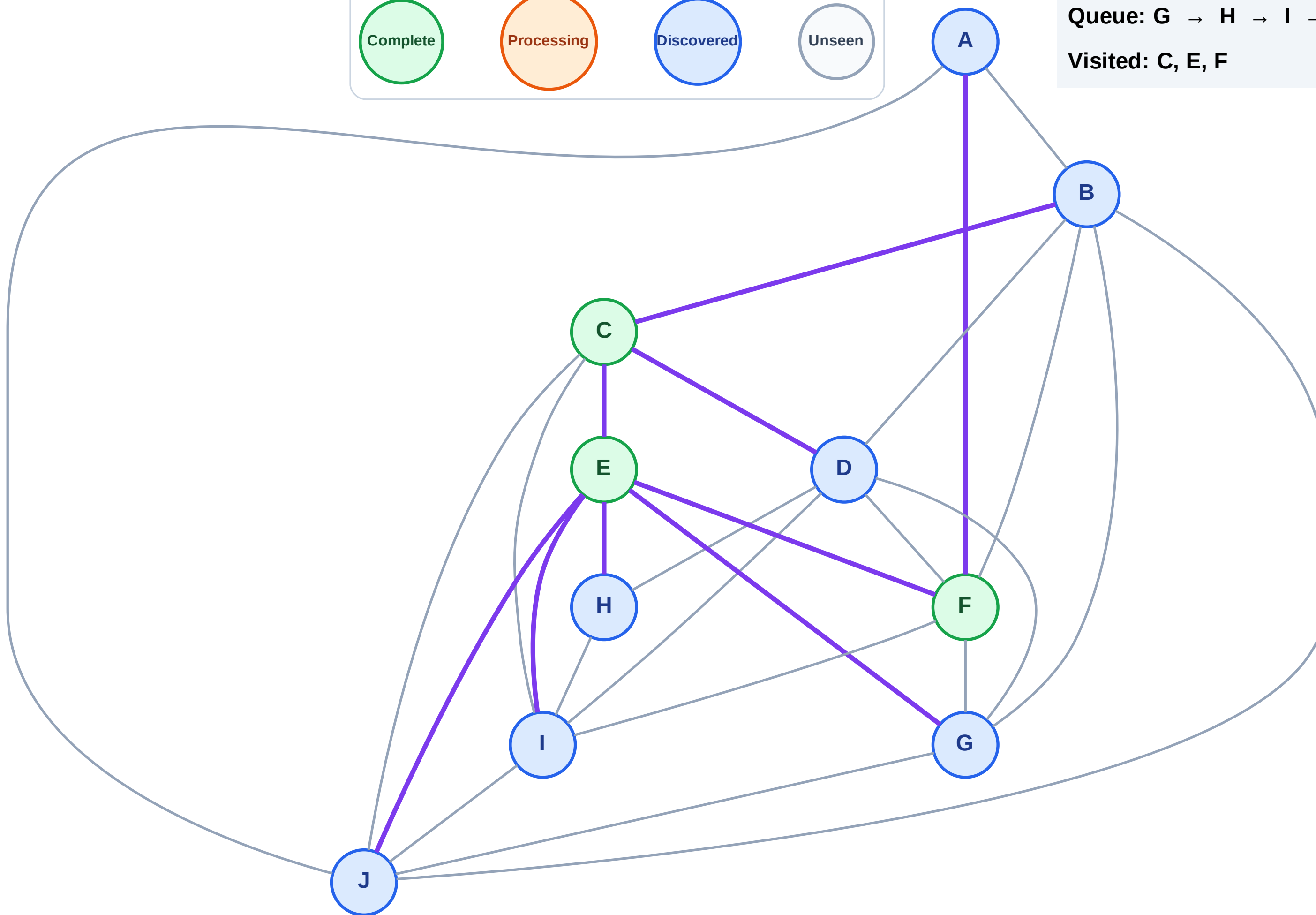
BFS Traversal

Step 7: Discover A from F

Legend



Queue: G → H → I → J → B → D → A
Visited: C, E, F



BFS Traversal

Step 8: Process node G

Legend

Complete

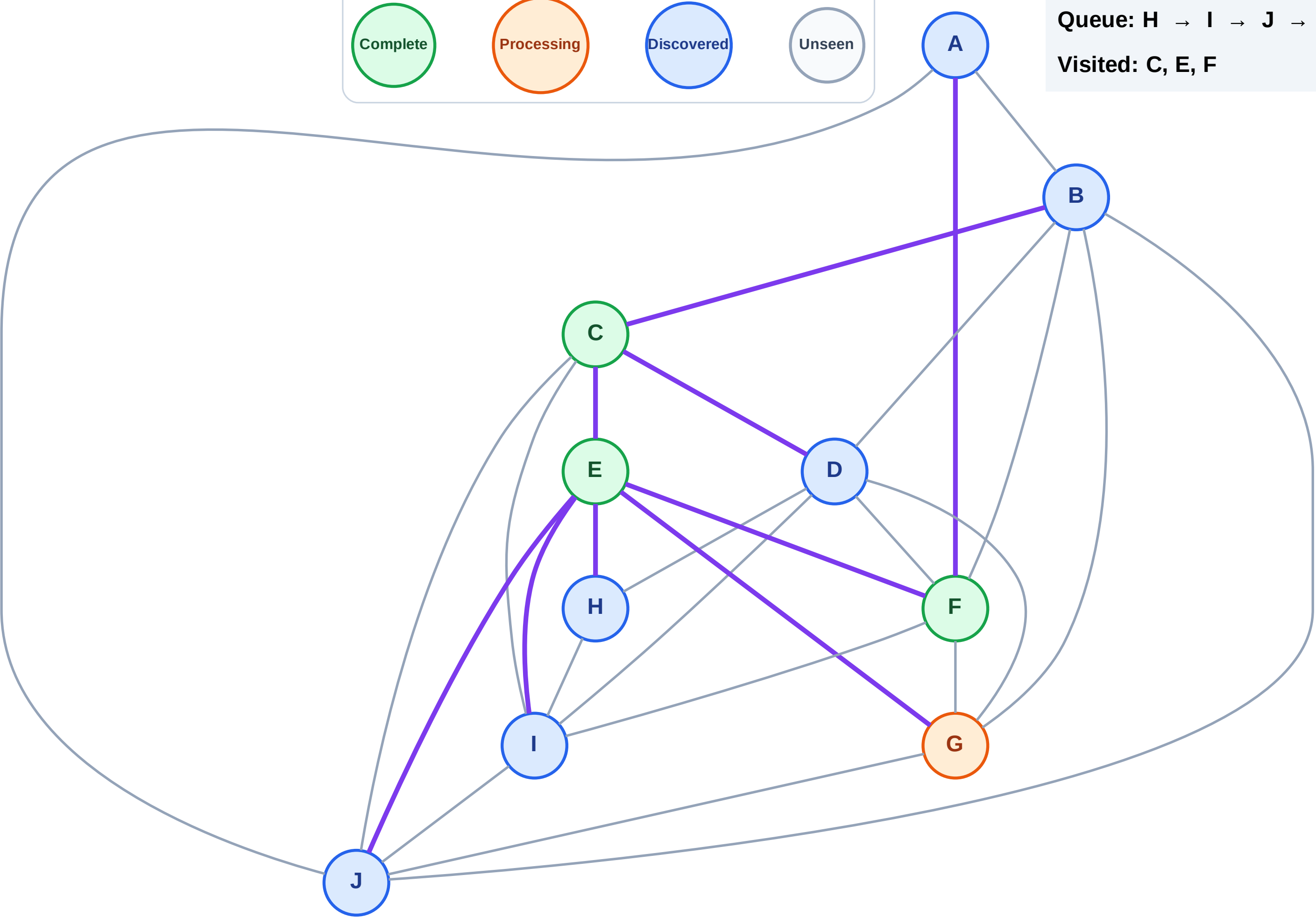
Processing

Discovered

Unseen

Queue: H → I → J → B → D → A

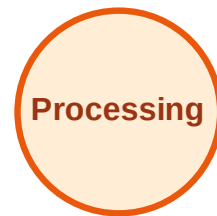
Visited: C, E, F



BFS Traversal

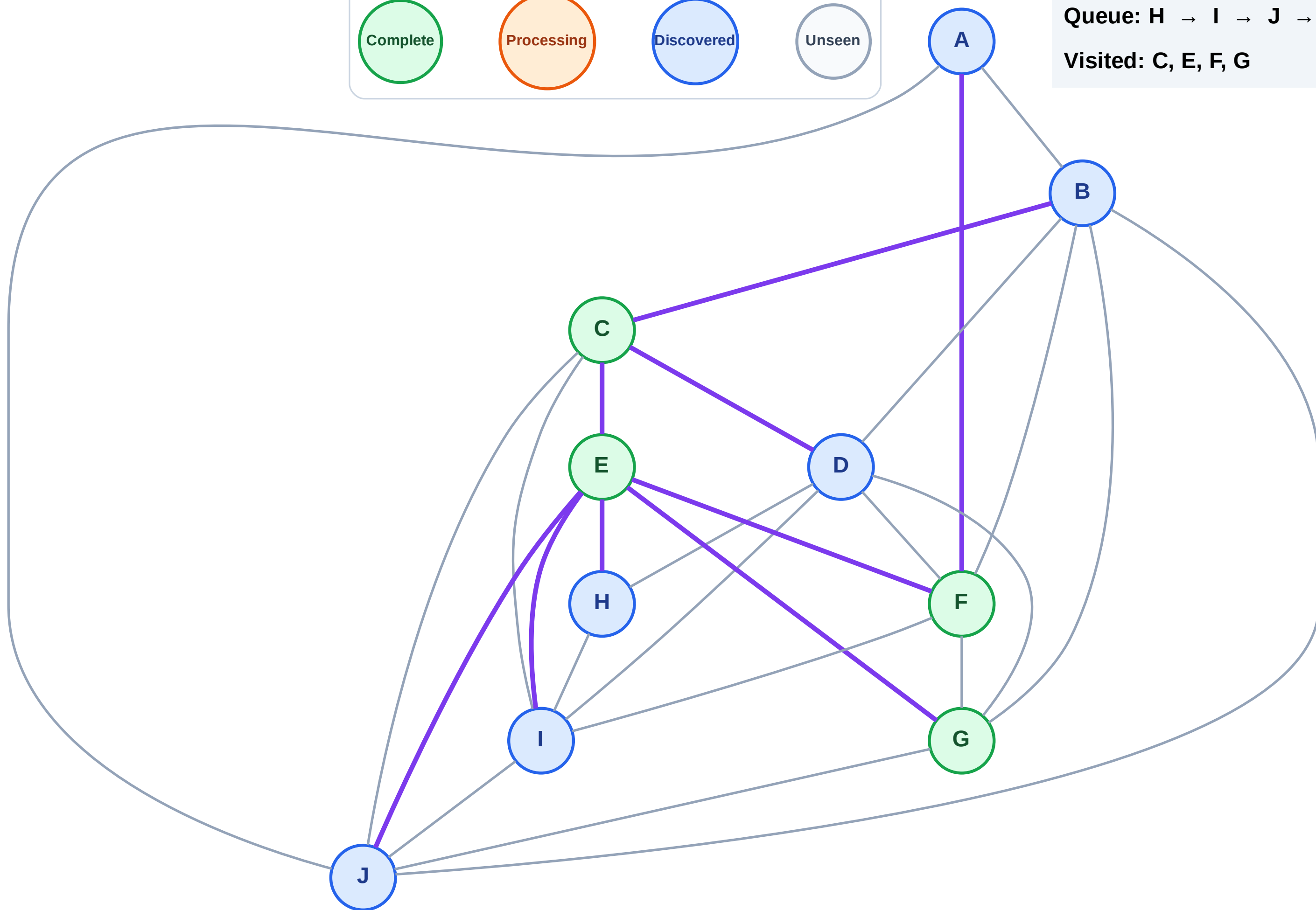
Step 9: No new neighbors from G

Legend



Queue: H → I → J → B → D → A

Visited: C, E, F, G



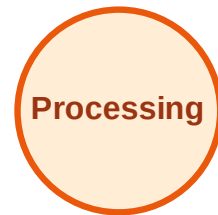
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



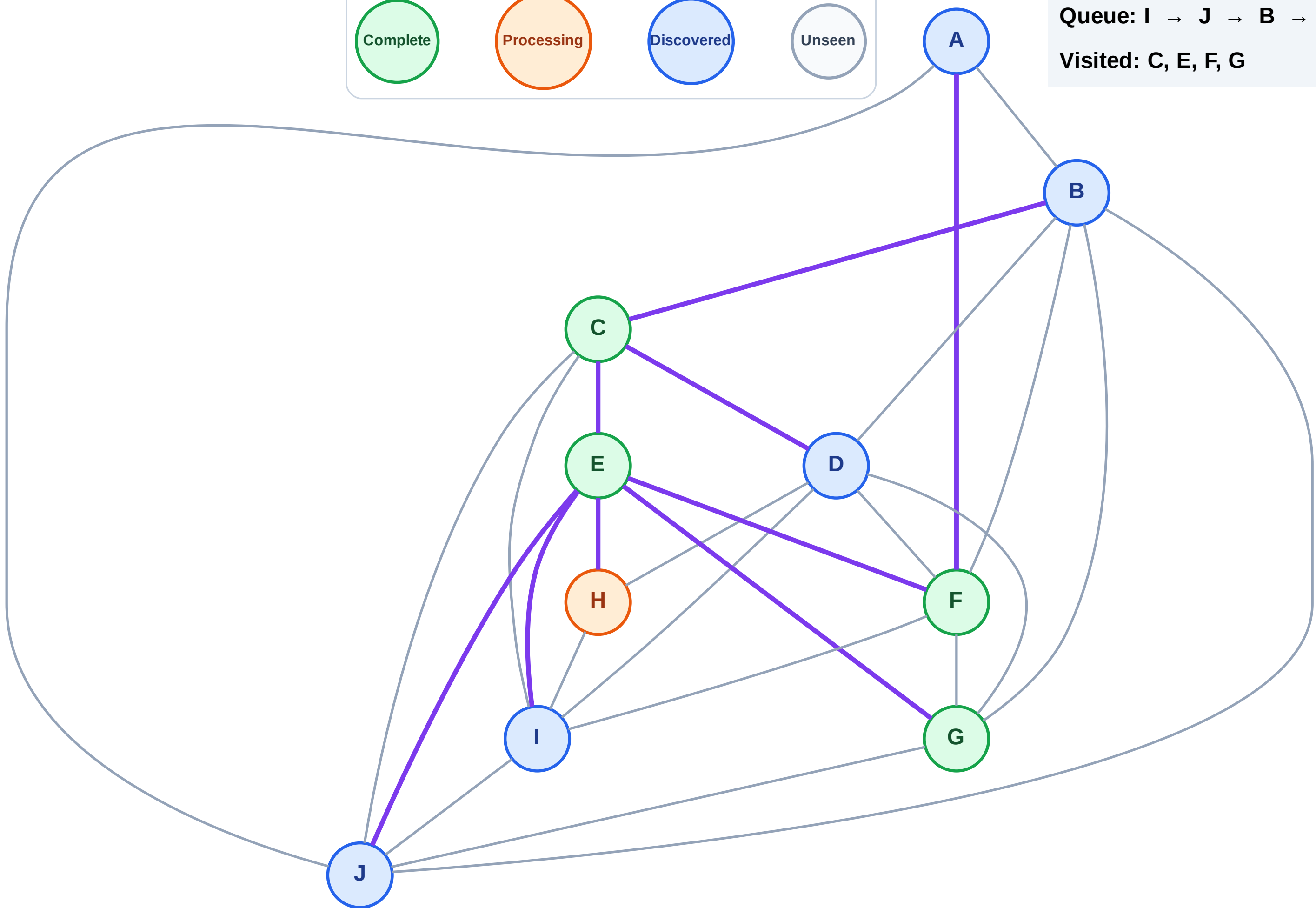
Discovered



Unseen

Queue: I → J → B → D → A

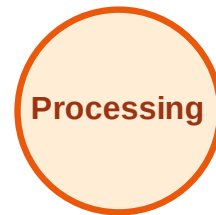
Visited: C, E, F, G



BFS Traversal

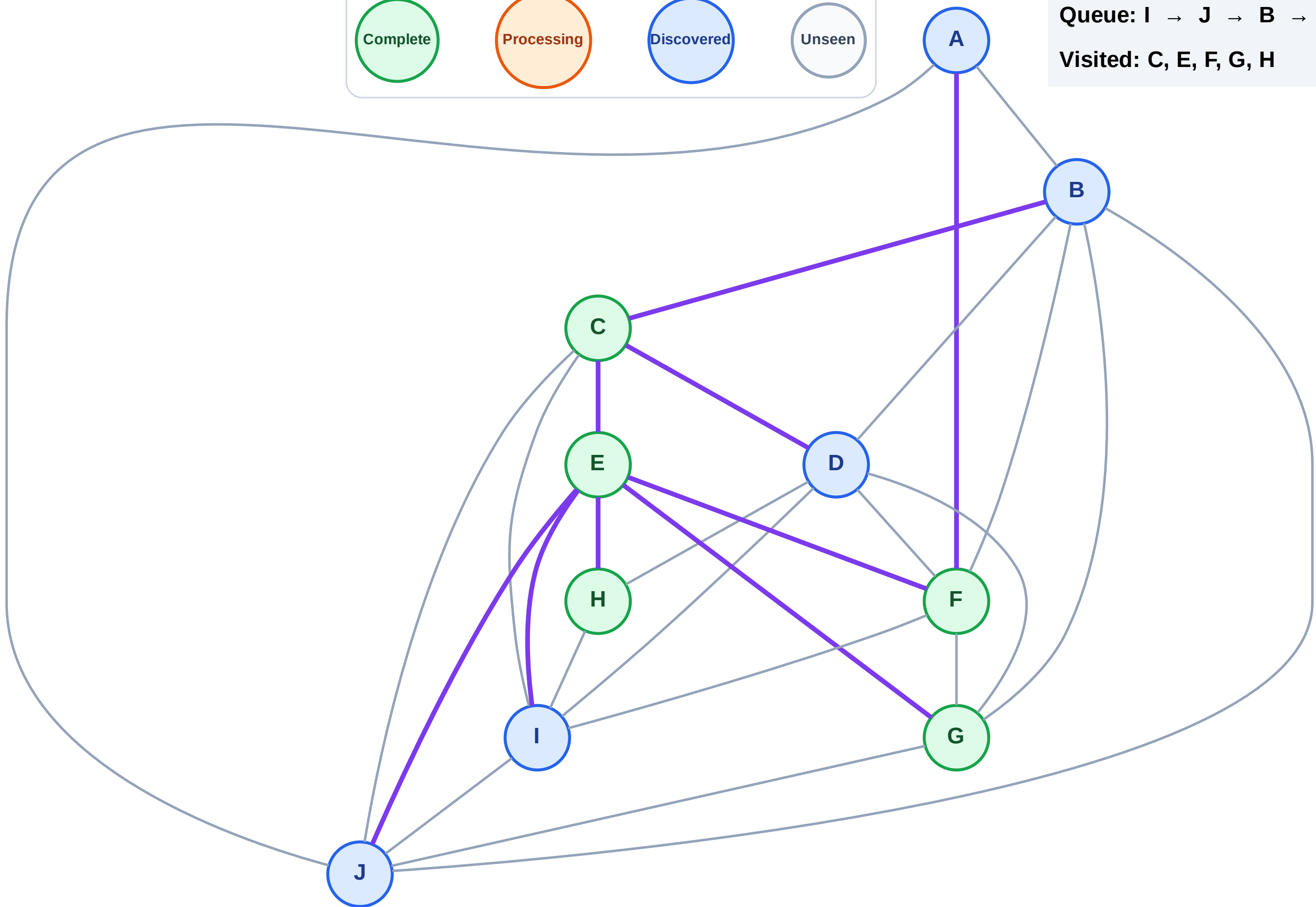
Step 11: No new neighbors from H

Legend



Queue: I → J → B → D → A

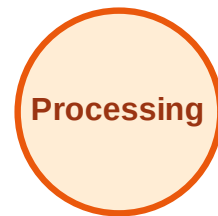
Visited: C, E, F, G, H



BFS Traversal

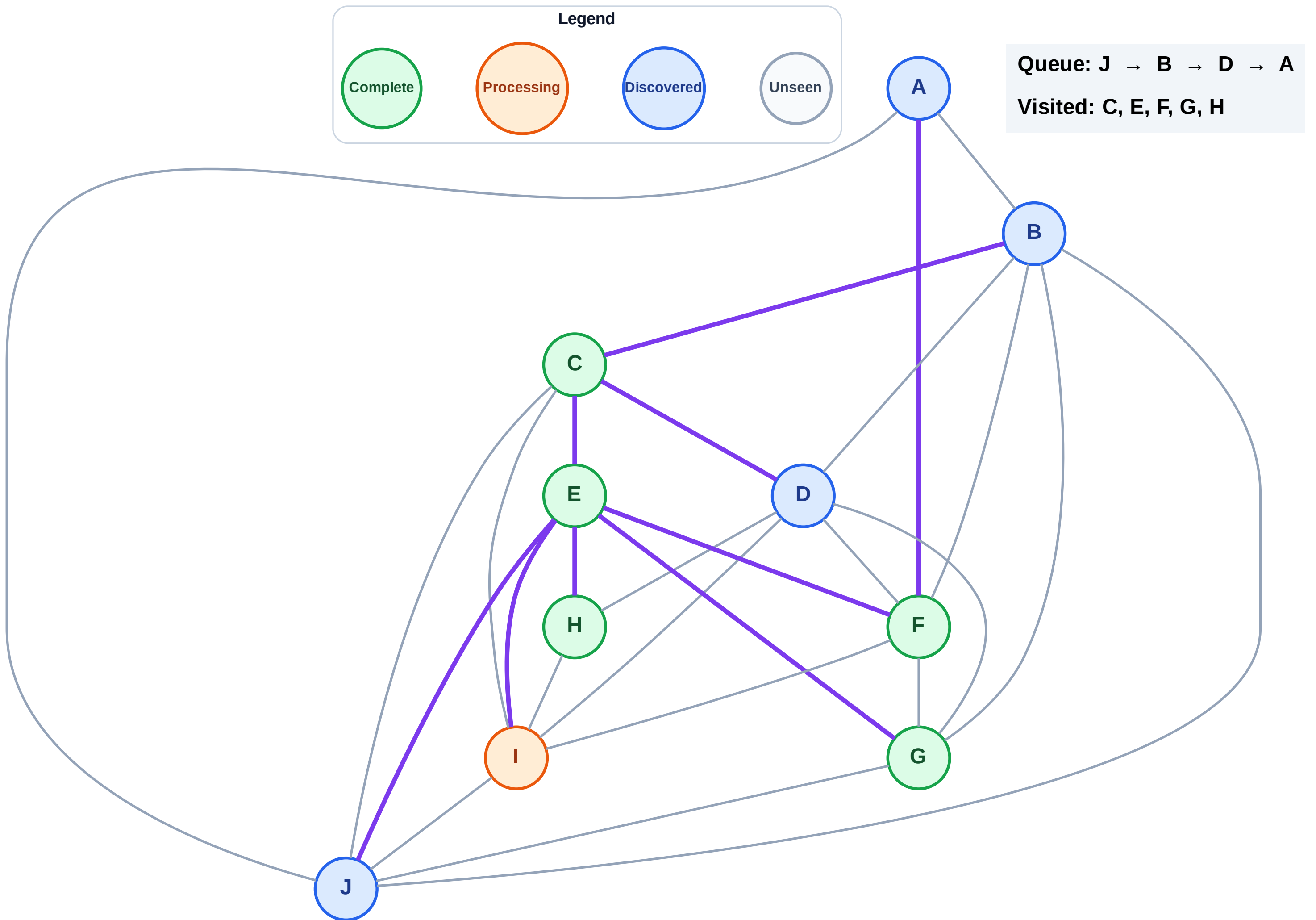
Step 12: Process node I

Legend



Queue: J → B → D → A

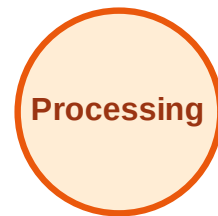
Visited: C, E, F, G, H



BFS Traversal

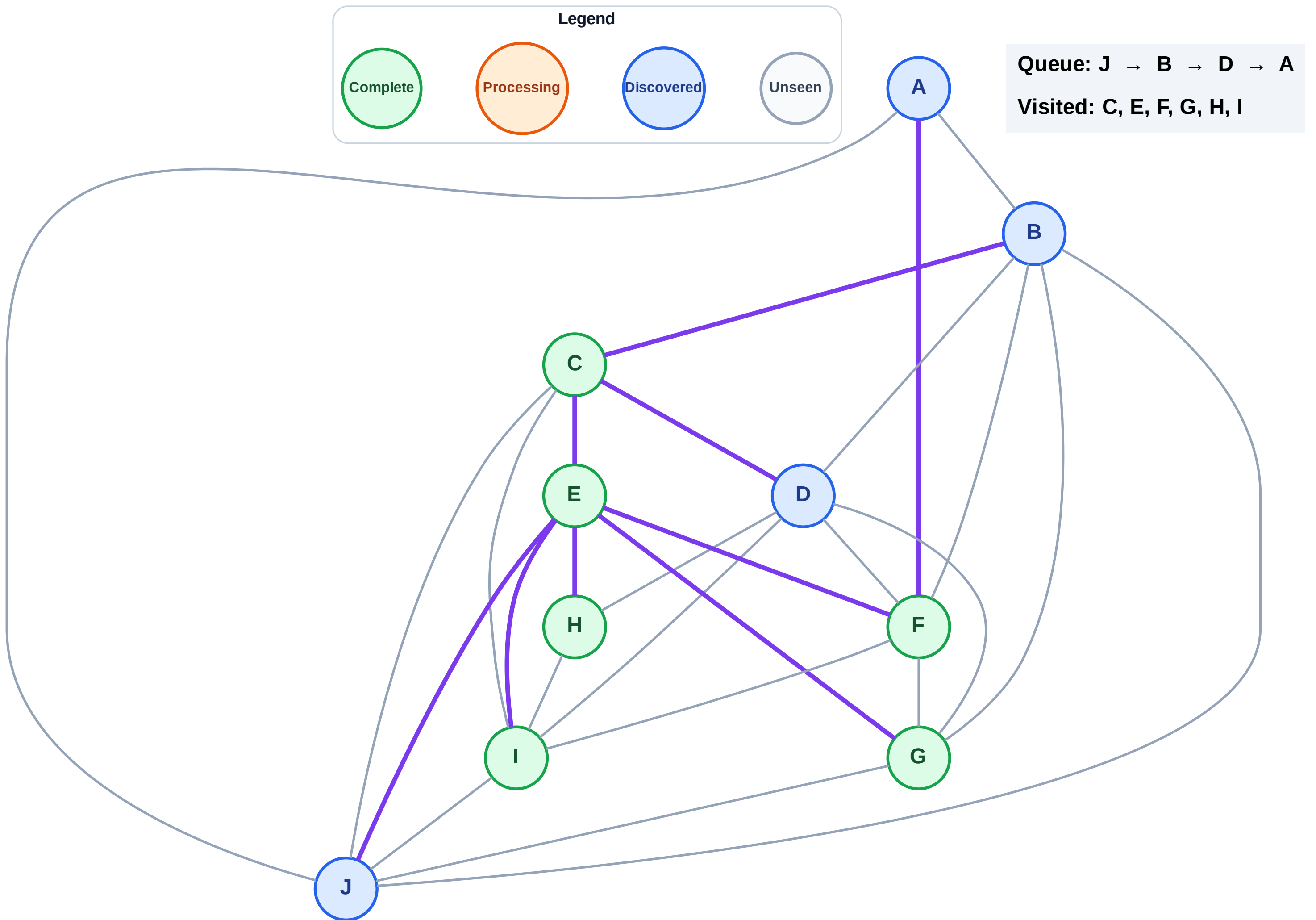
Step 13: No new neighbors from I

Legend



Queue: J → B → D → A

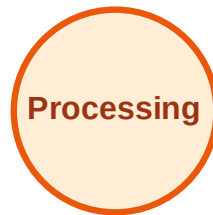
Visited: C, E, F, G, H, I



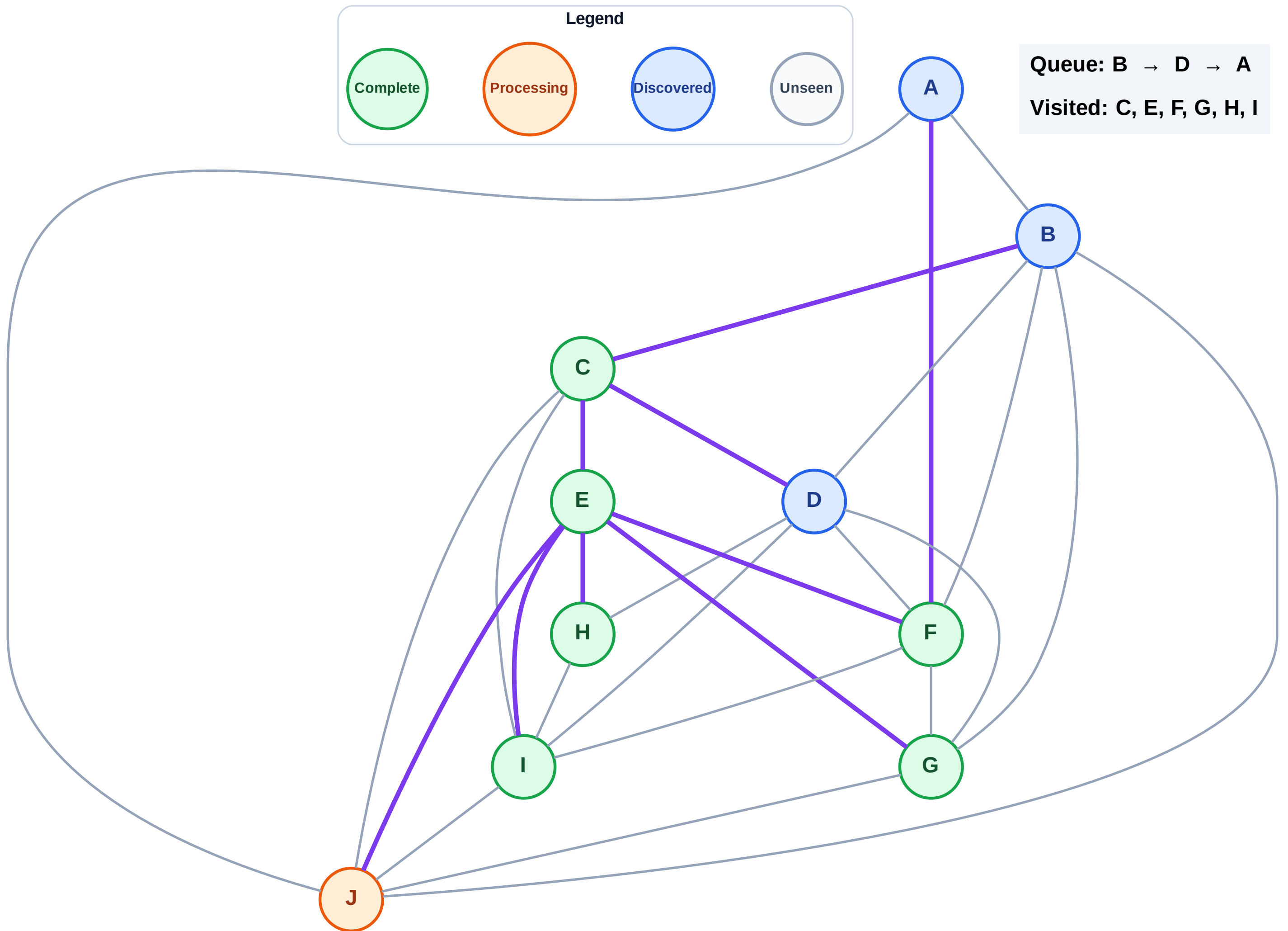
BFS Traversal

Step 14: Process node J

Legend



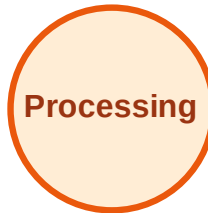
Queue: B → D → A
Visited: C, E, F, G, H, I



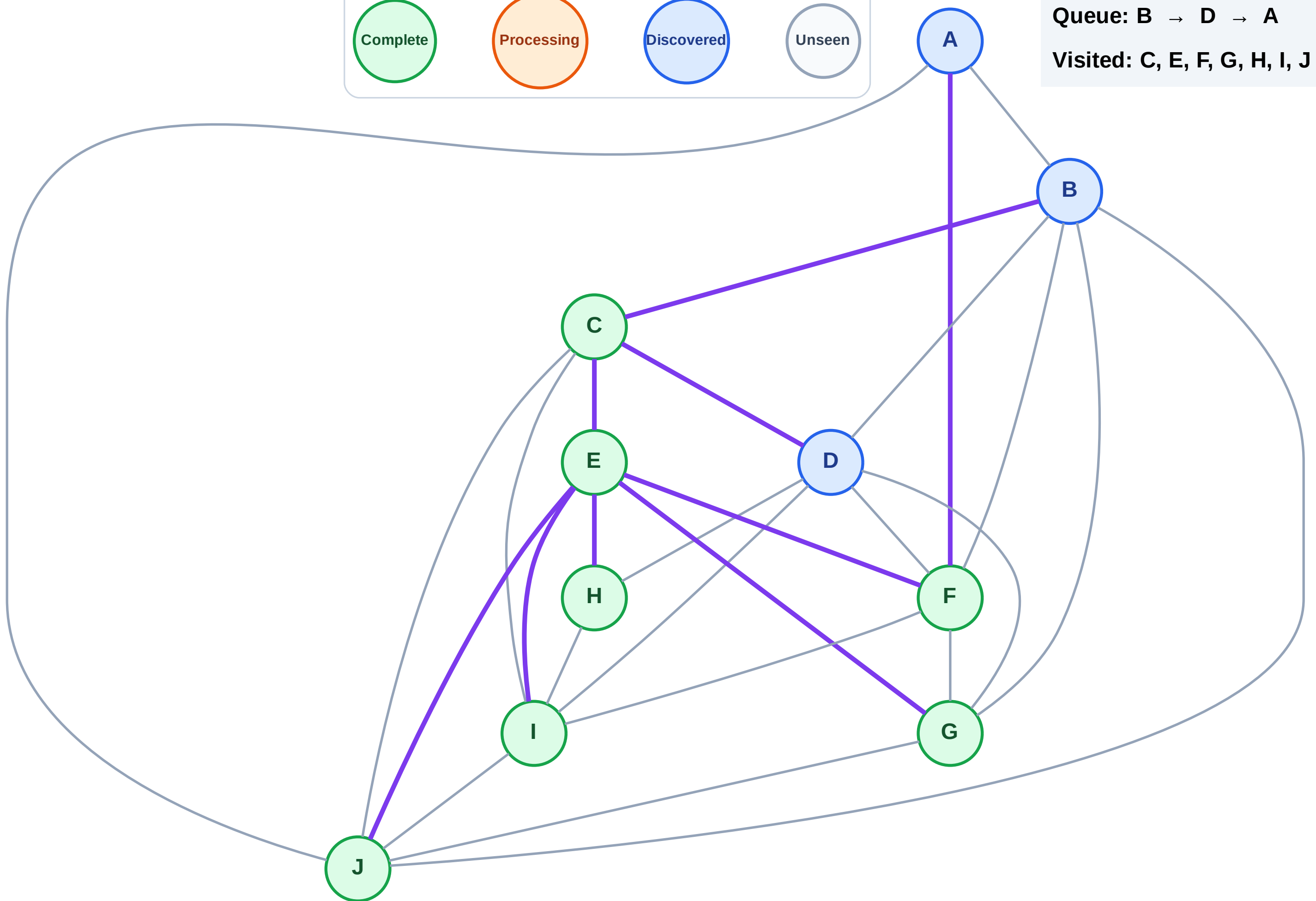
BFS Traversal

Step 15: No new neighbors from J

Legend



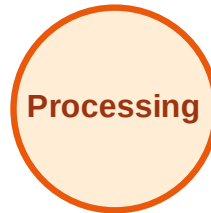
Queue: B → D → A
Visited: C, E, F, G, H, I, J



BFS Traversal

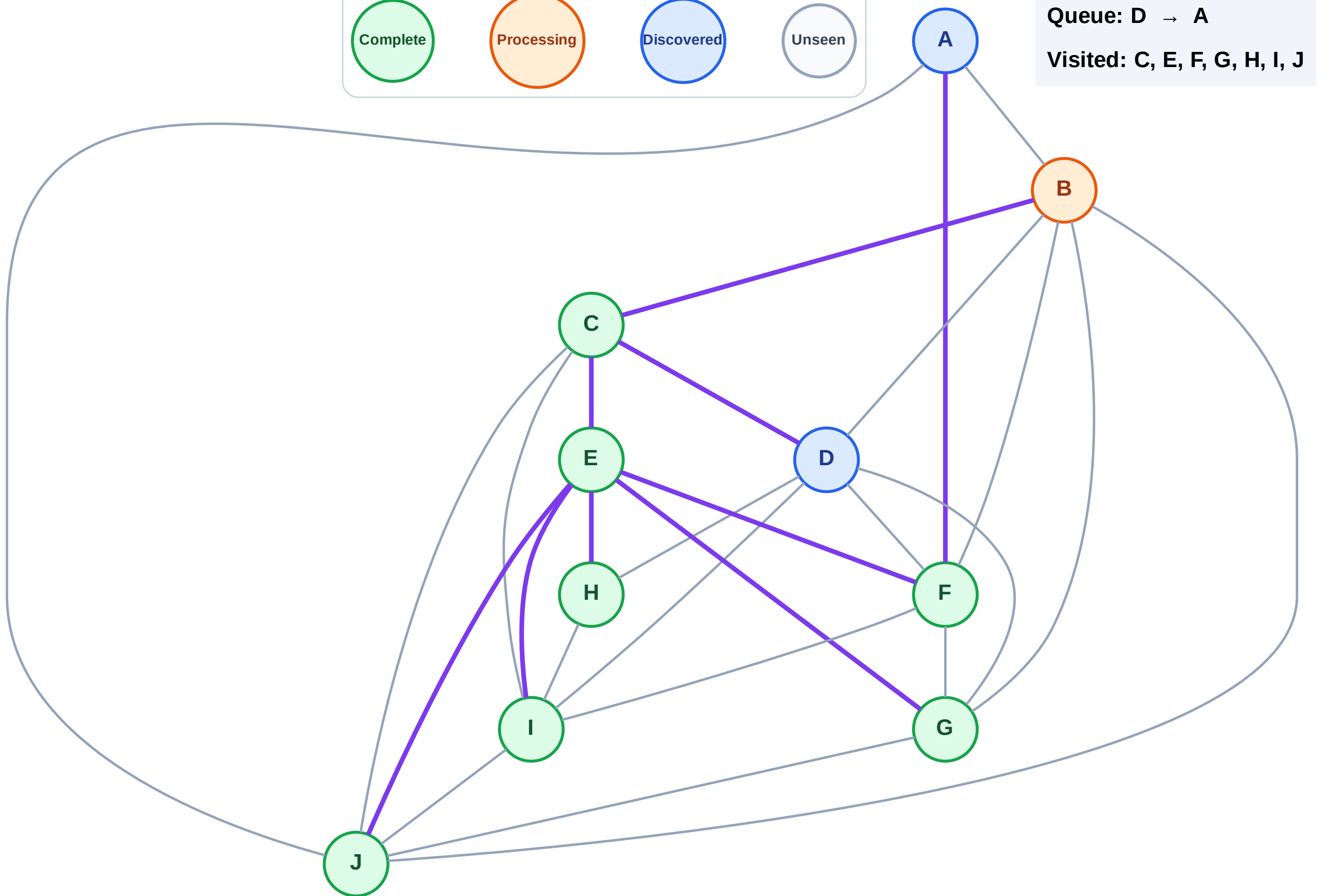
Step 16: Process node B

Legend



Queue: D → A

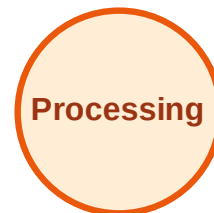
Visited: C, E, F, G, H, I, J



BFS Traversal

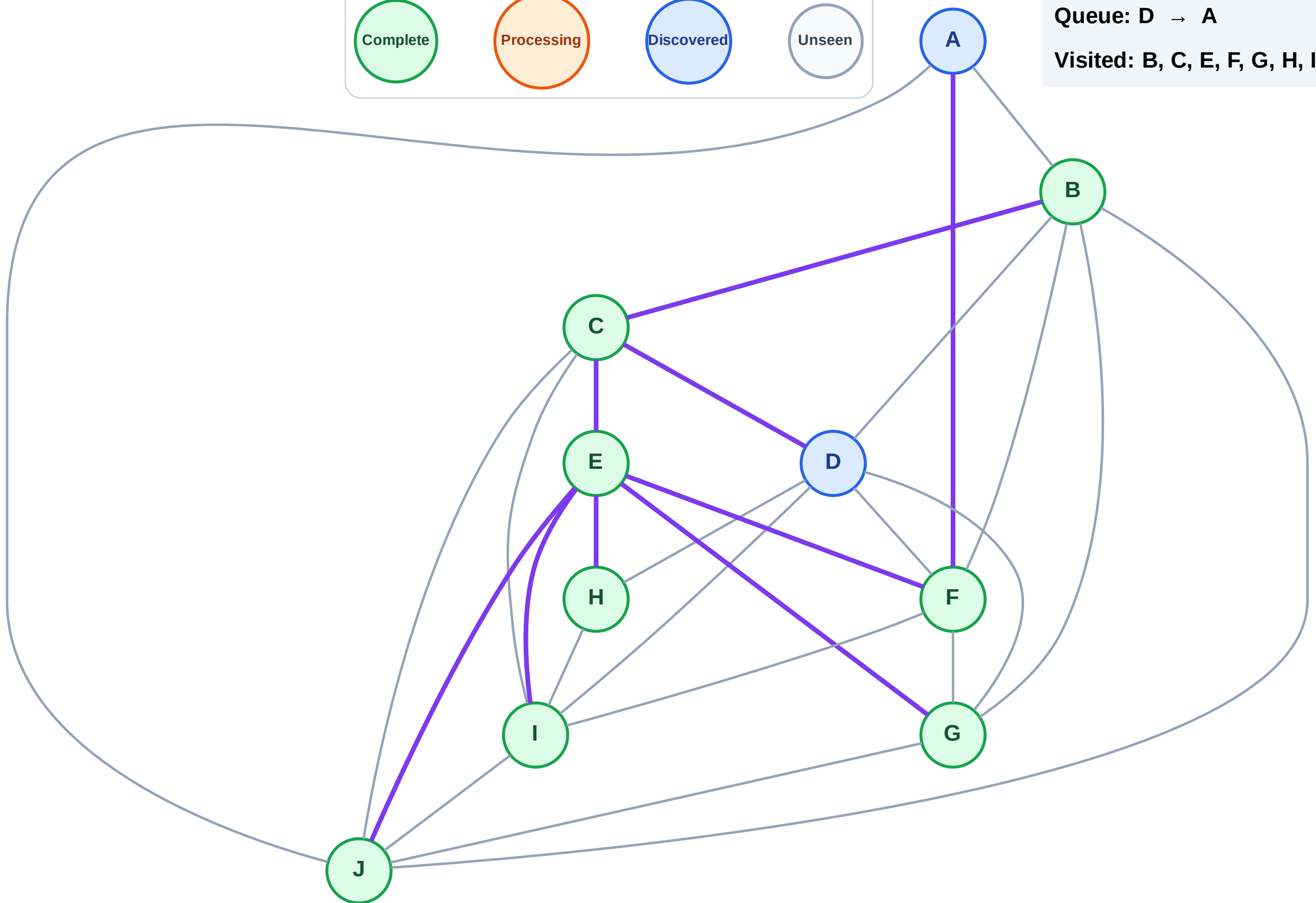
Step 17: No new neighbors from B

Legend



Queue: D → A

Visited: B, C, E, F, G, H, I, J



BFS Traversal

Step 18: Process node D

Legend

Complete

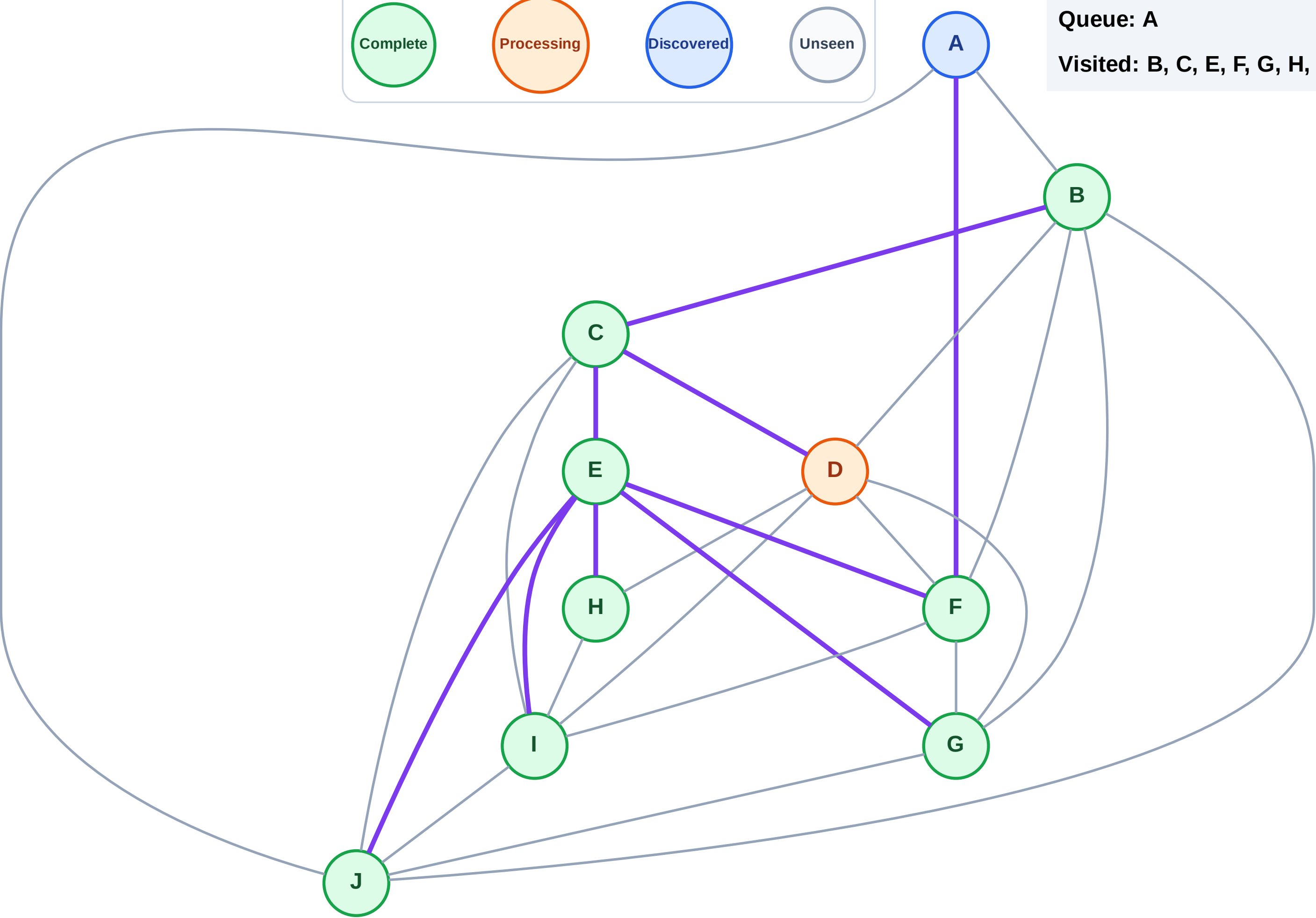
Processing

Discovered

Unseen

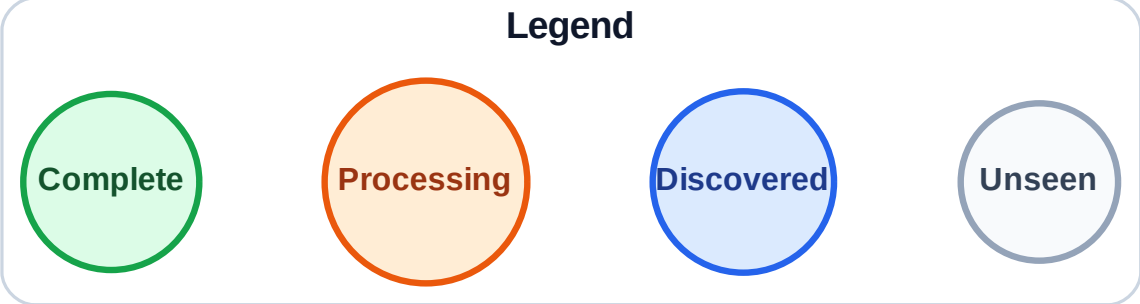
Queue: A

Visited: B, C, E, F, G, H, I, J



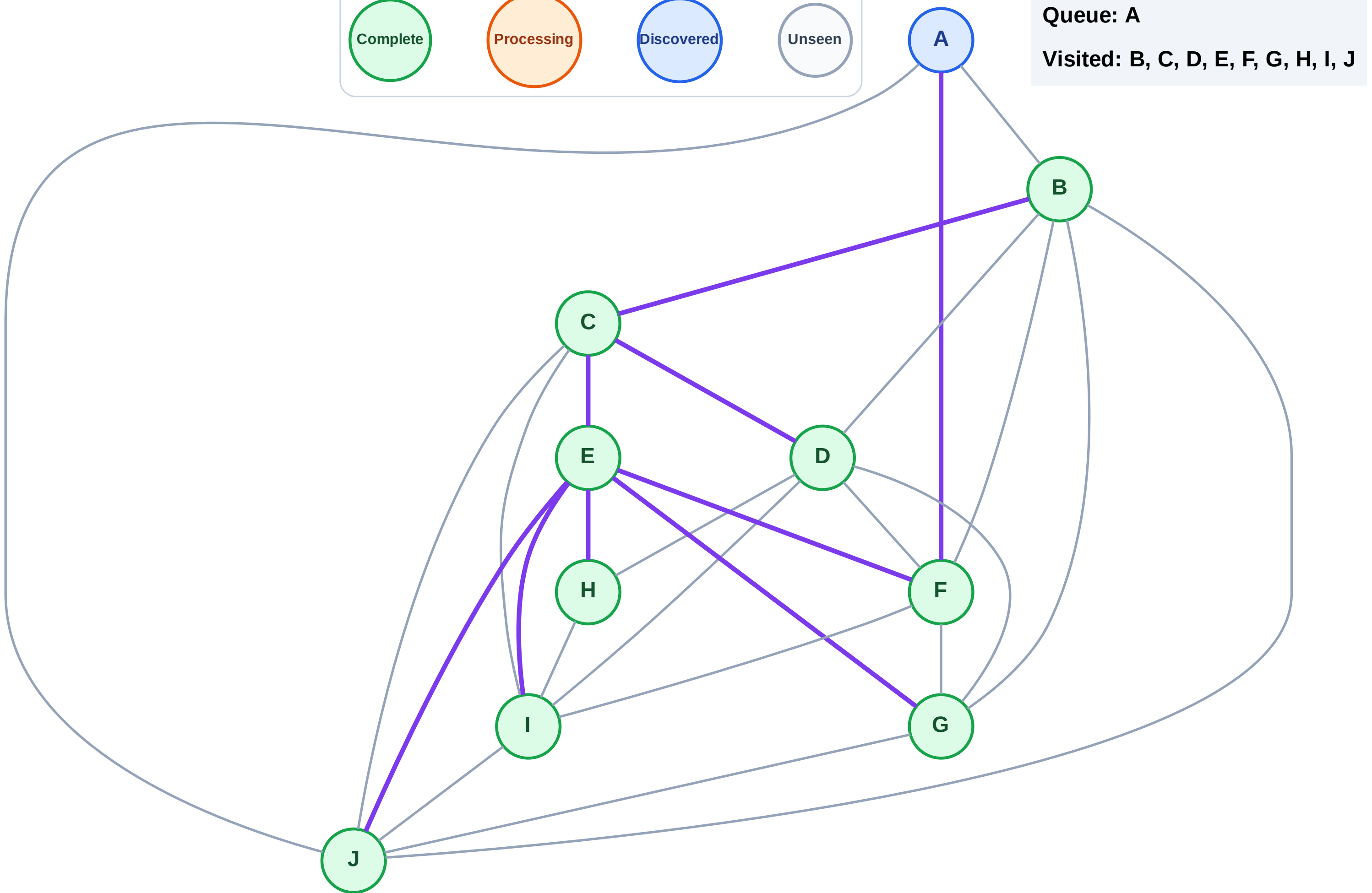
BFS Traversal

Step 19: No new neighbors from D



Queue: A

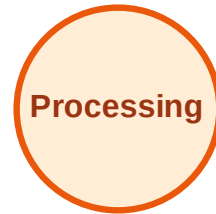
Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

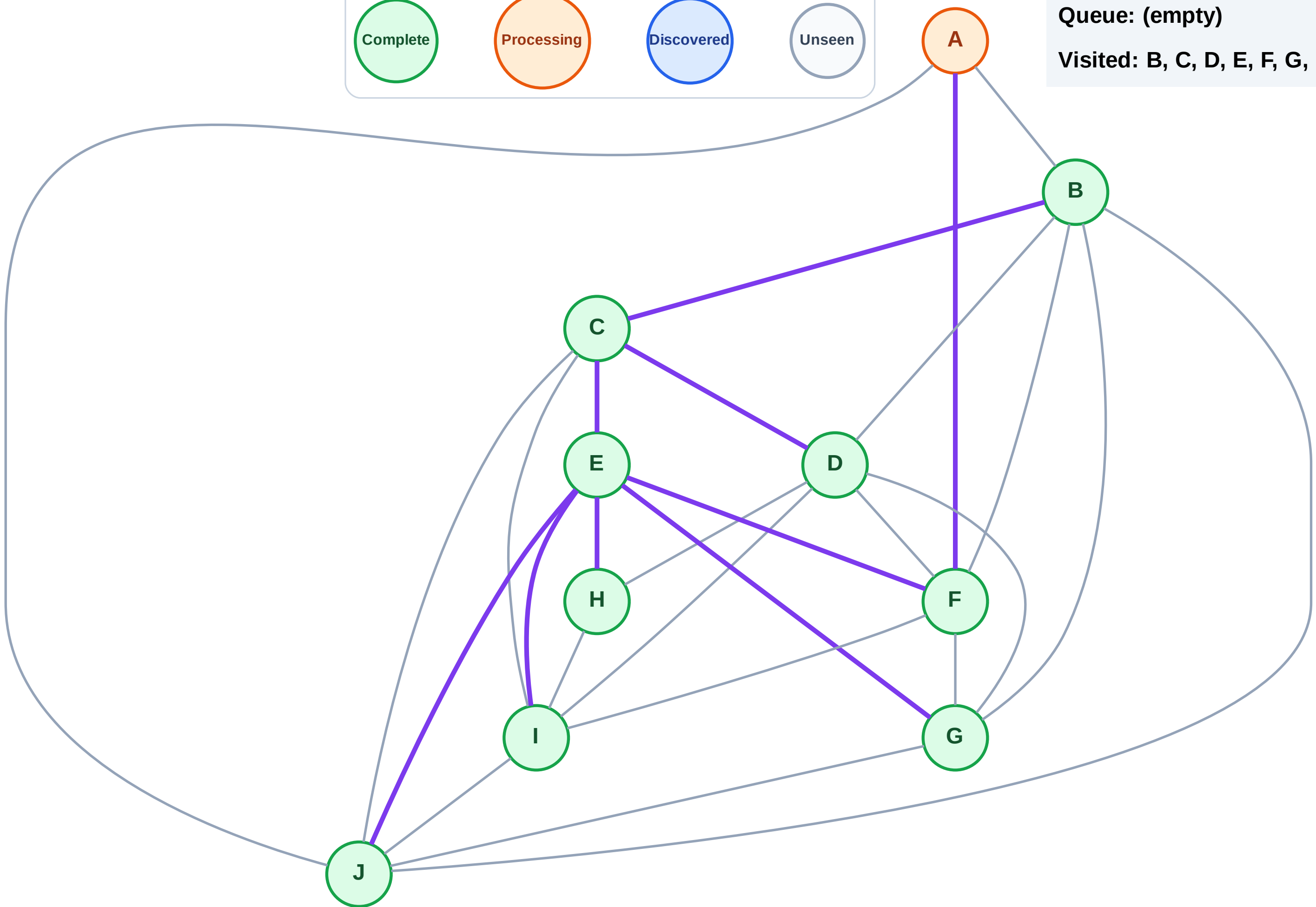
Step 20: Process node A

Legend



Queue: (empty)

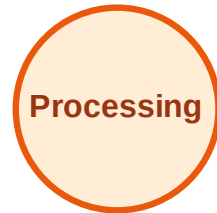
Visited: B, C, D, E, F, G, H, I, J



BFS Traversal

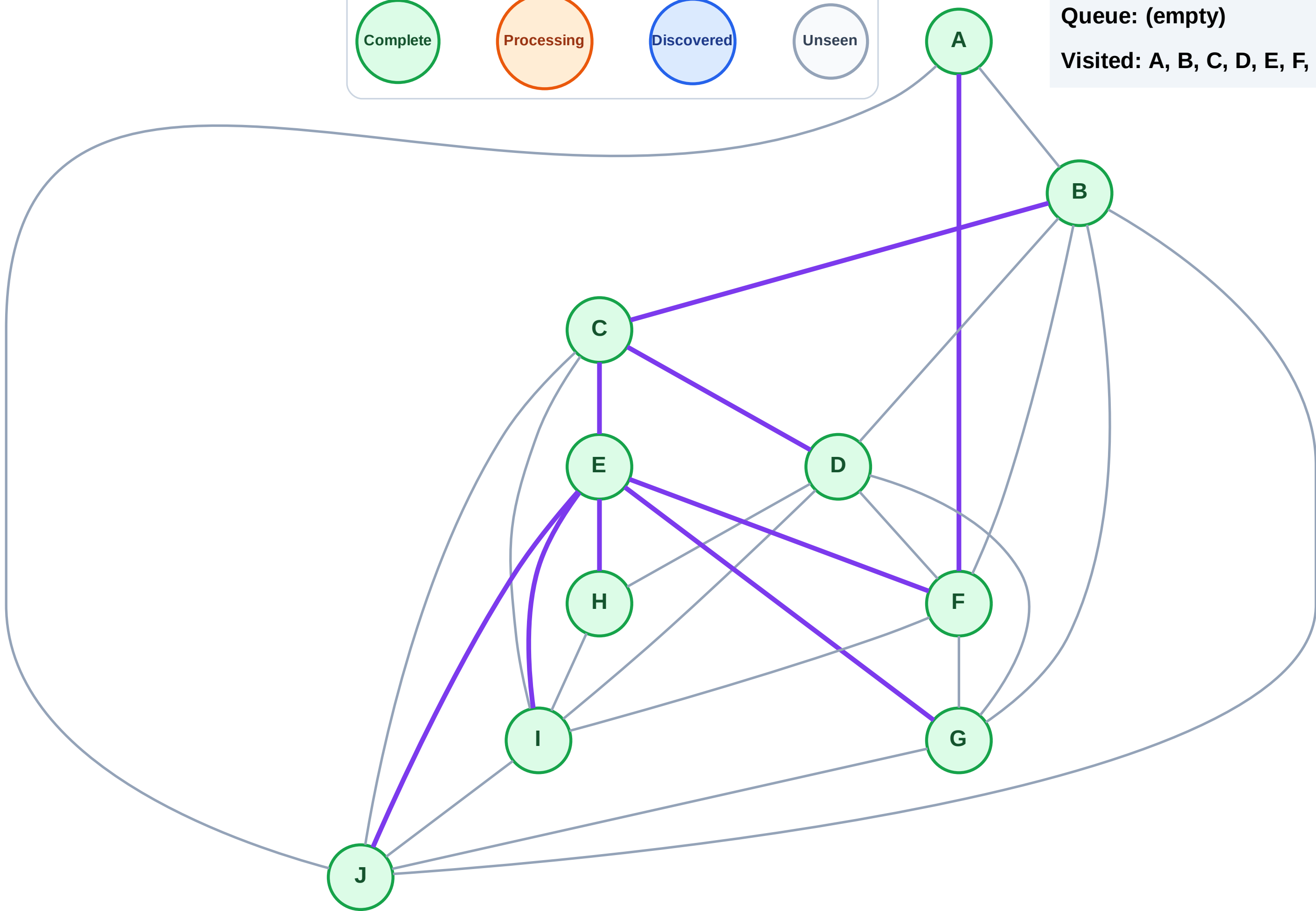
Step 21: No new neighbors from A

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



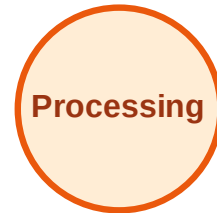
DFS Traversal

Step 1: Start at E

Legend



Complete



Processing



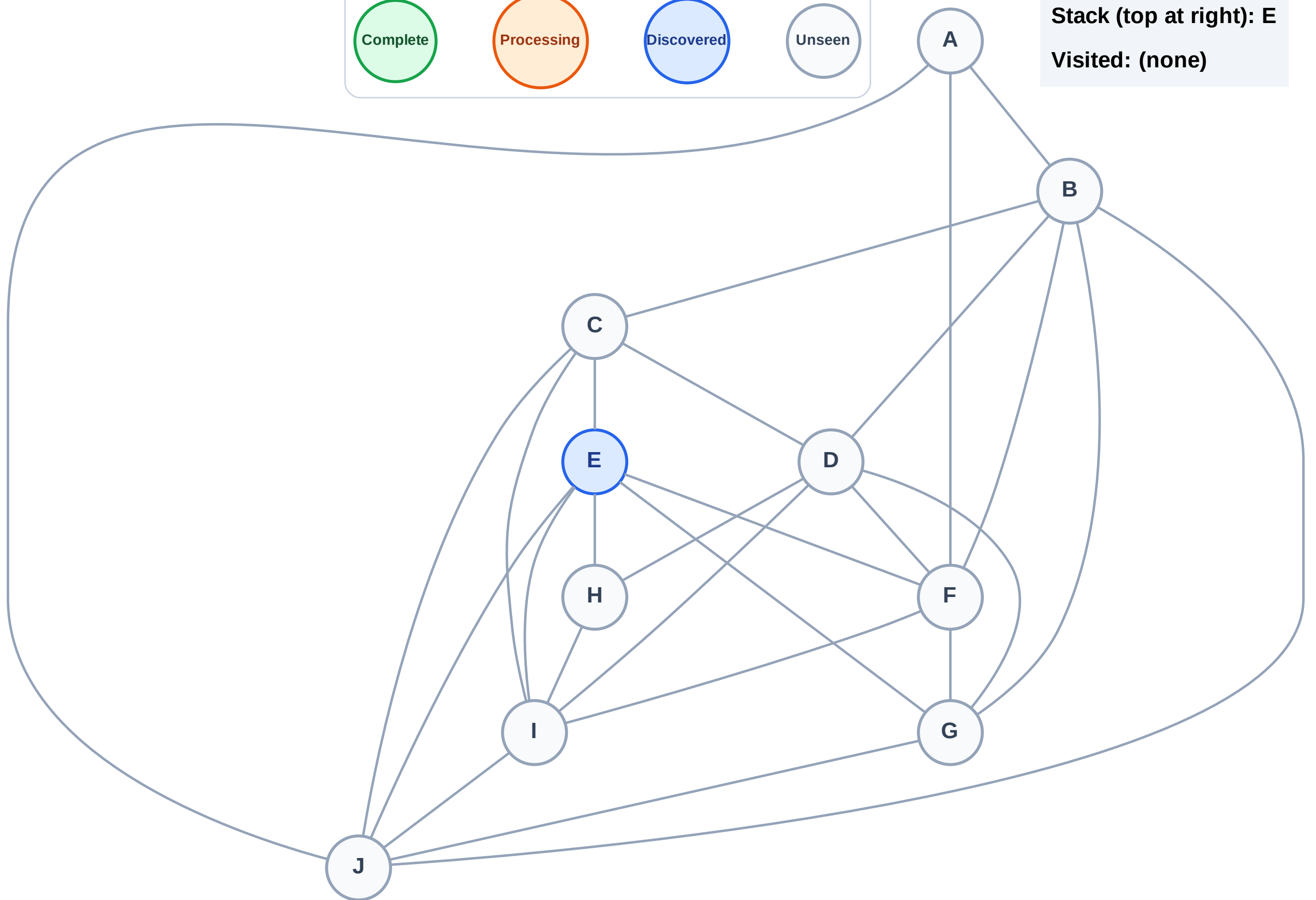
Discovered



Unseen

Stack (top at right): E

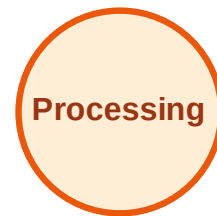
Visited: (none)



DFS Traversal

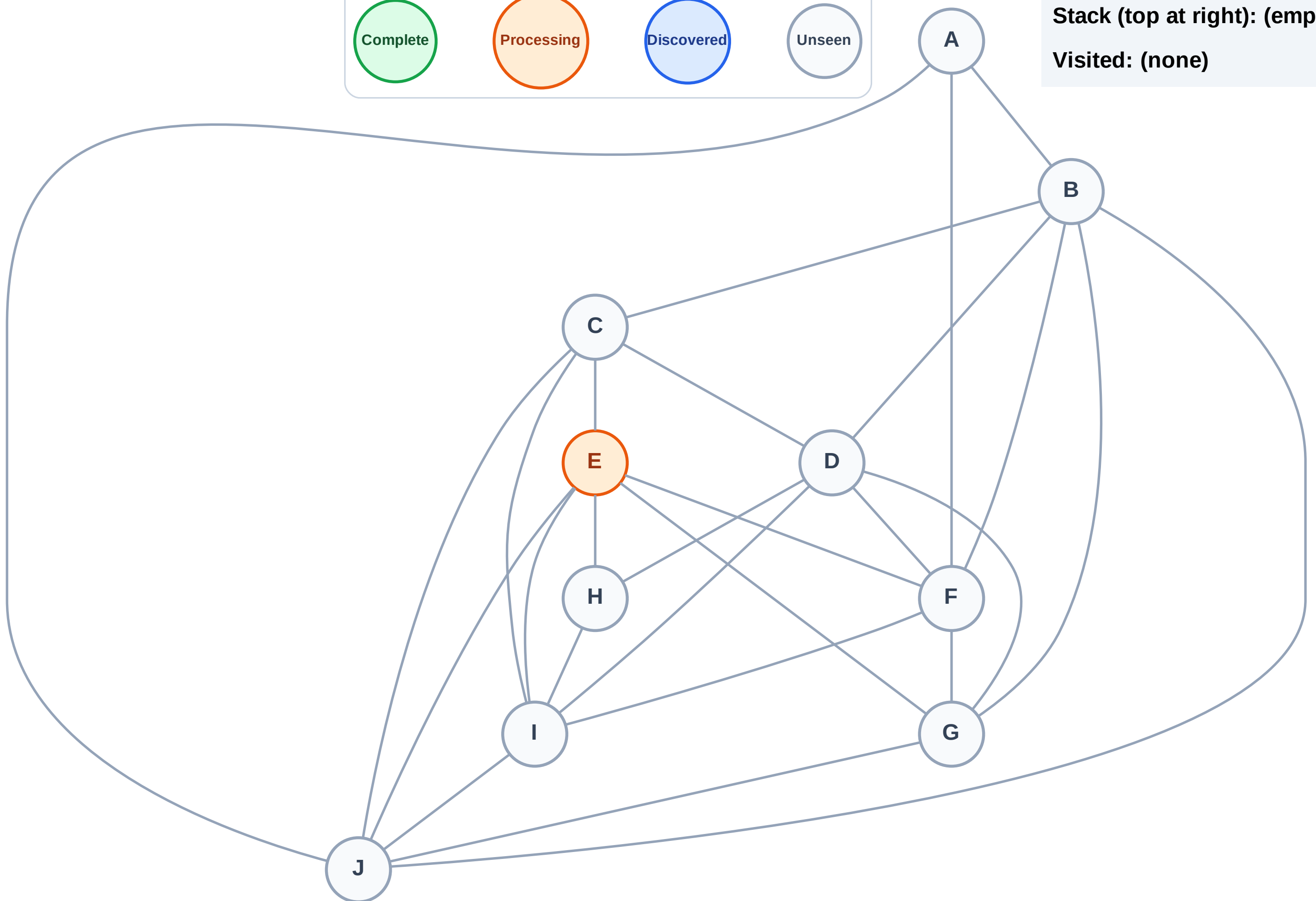
Step 2: Process node E

Legend



Stack (top at right): (empty)

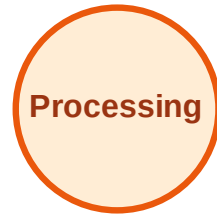
Visited: (none)



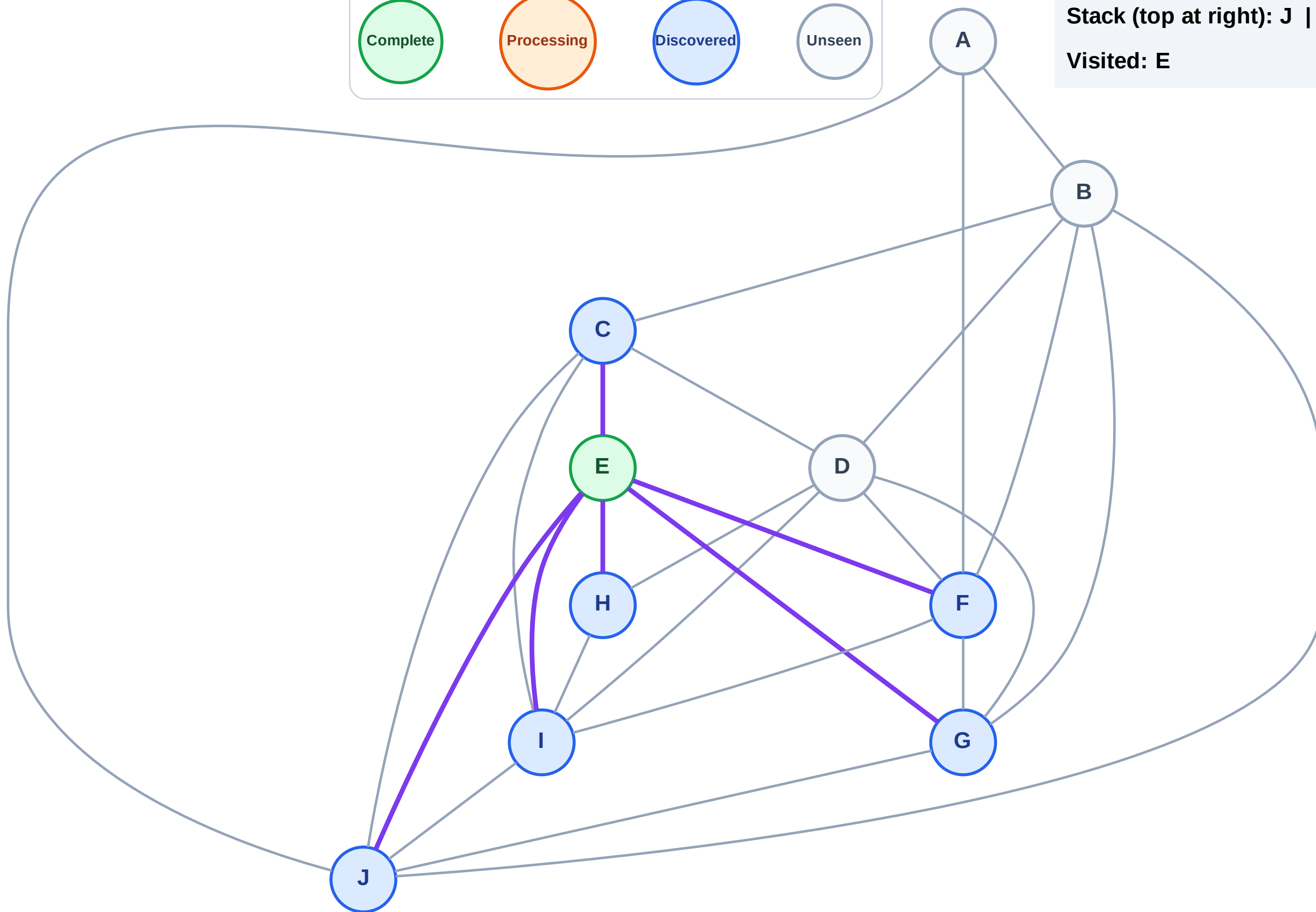
DFS Traversal

Step 3: Discover J, I, H, G, F, C from E

Legend



Stack (top at right): J | I | H | G | F | C
Visited: E



DFS Traversal

Step 4: Process node C

Legend

Complete

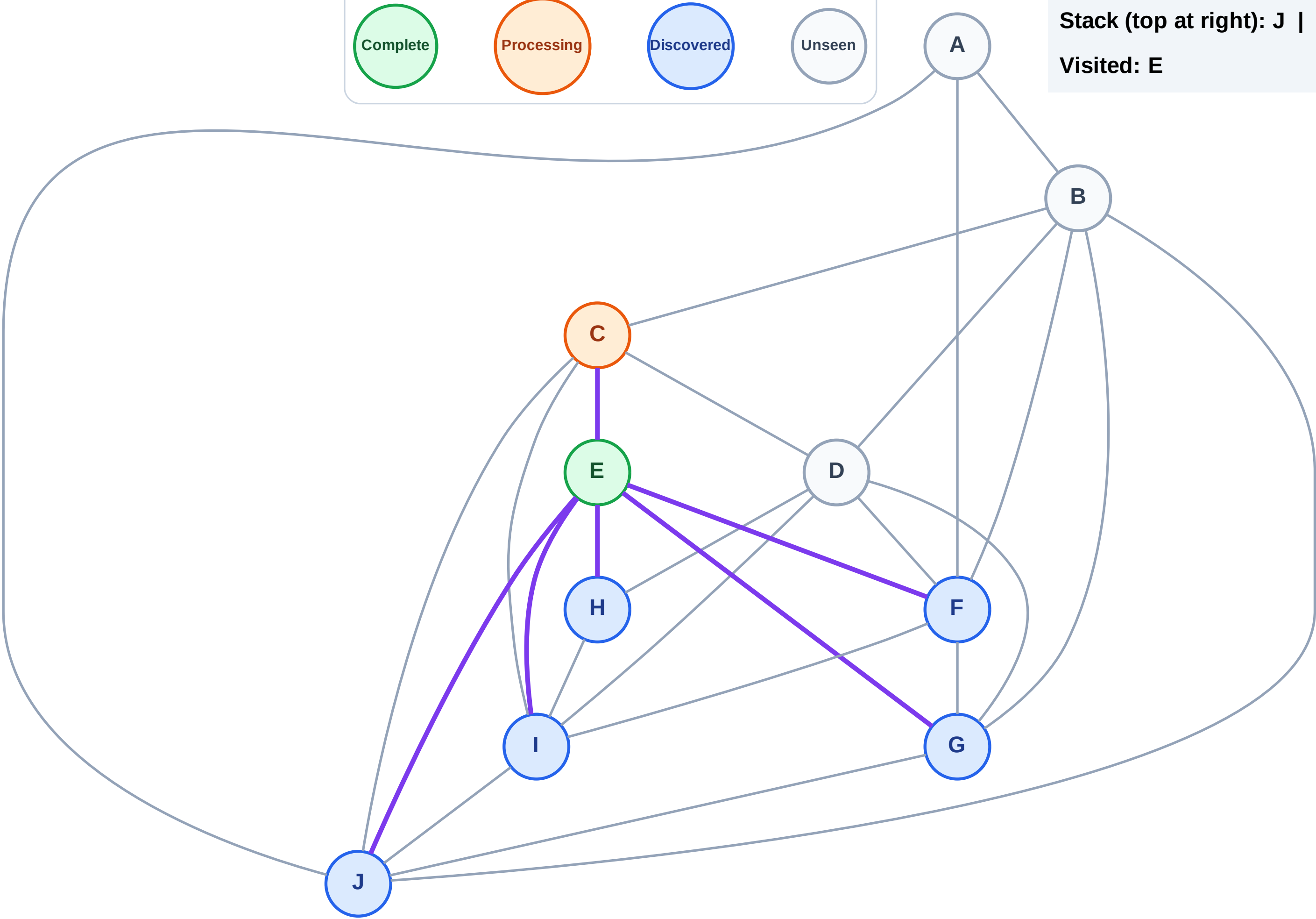
Processing

Discovered

Unseen

Stack (top at right): J | I | H | G | F

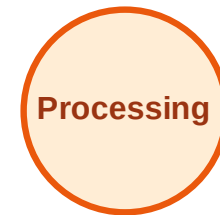
Visited: E



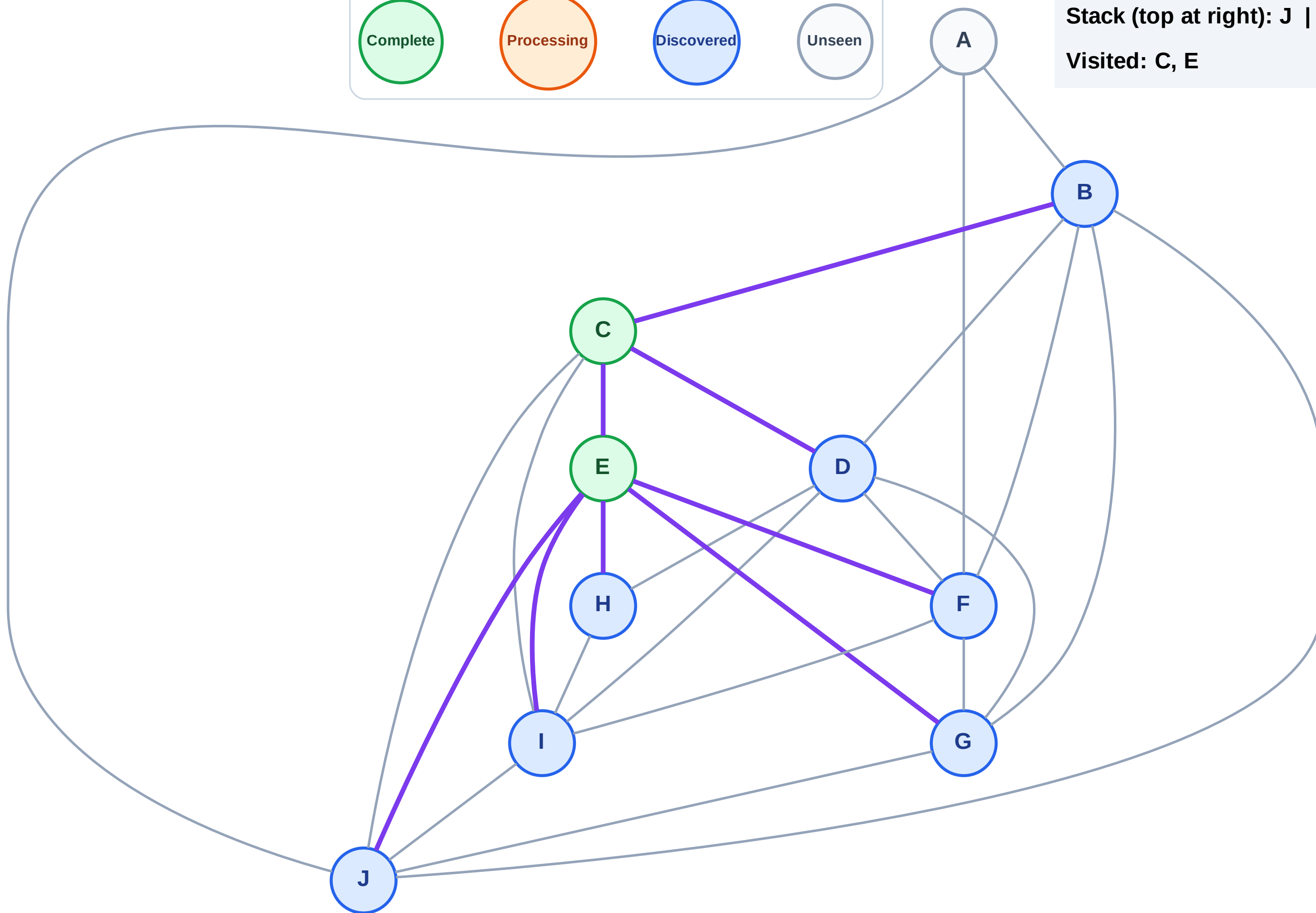
DFS Traversal

Step 5: Discover D, B from C

Legend



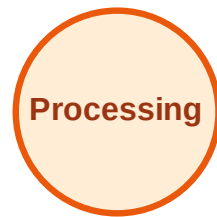
Stack (top at right): J | I | H | G | F | D | B
Visited: C, E



DFS Traversal

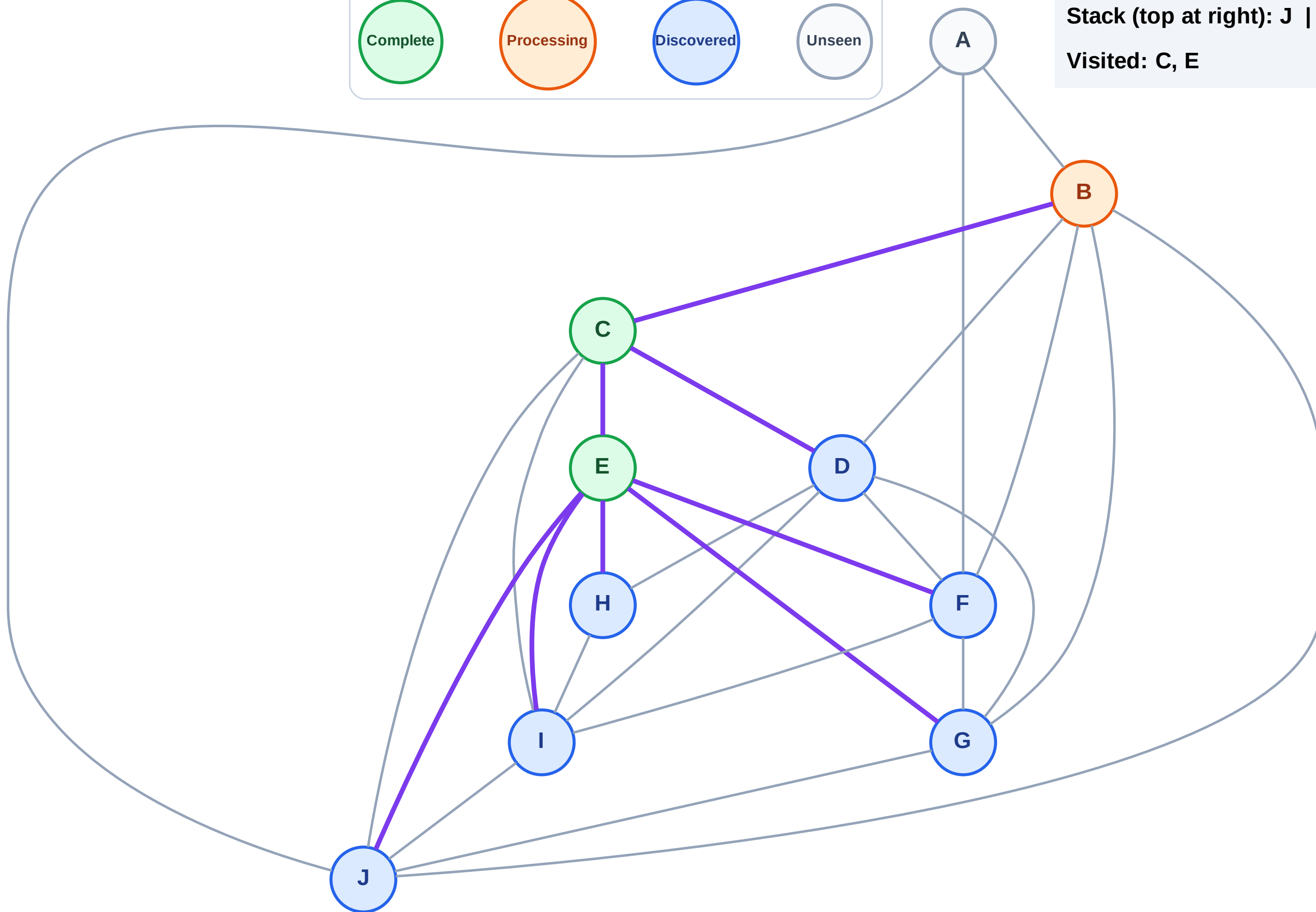
Step 6: Process node B

Legend



Stack (top at right): J | I | H | G | F | D

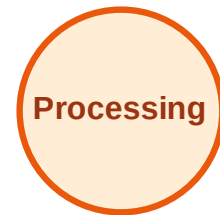
Visited: C, E



DFS Traversal

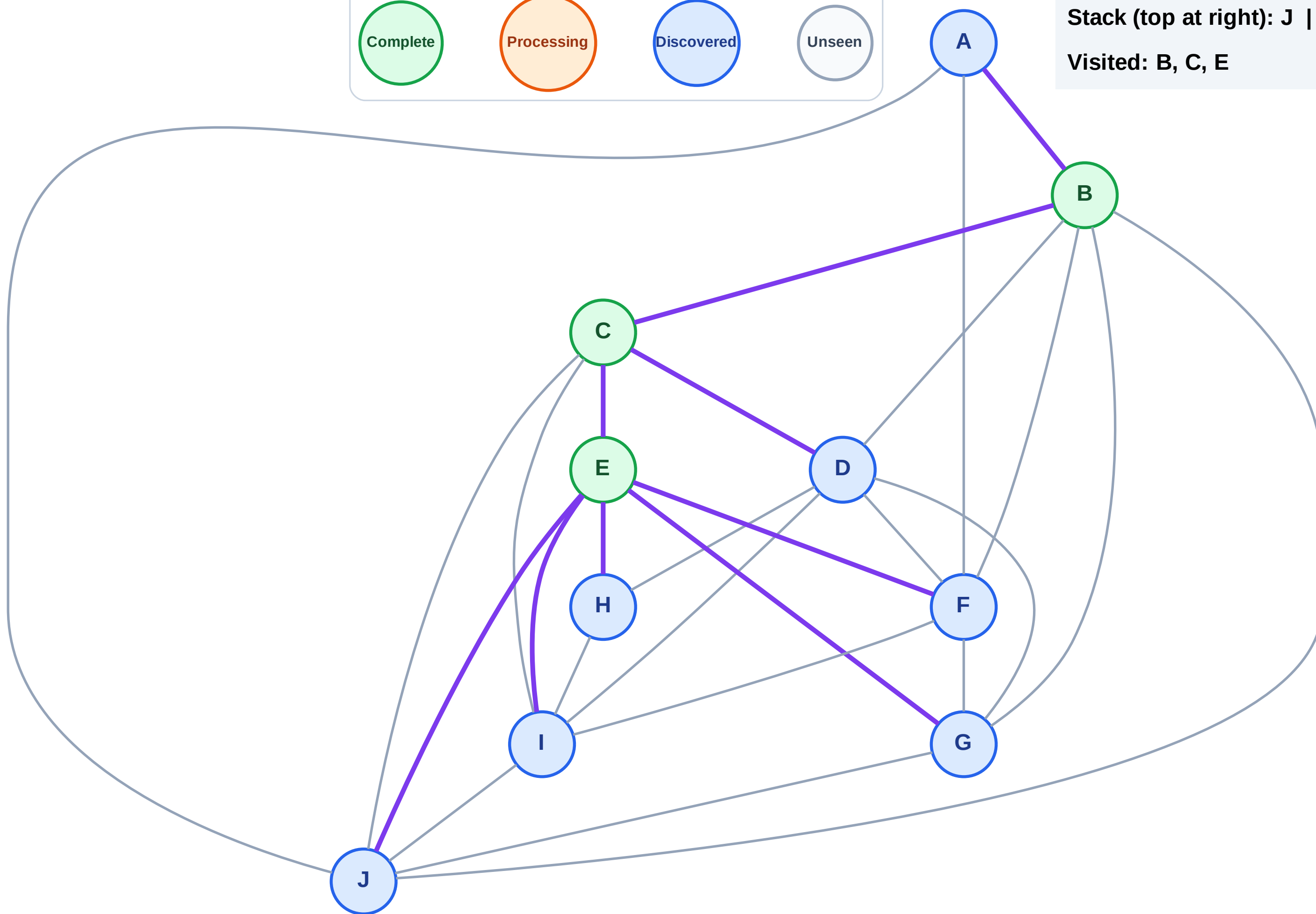
Step 7: Discover A from B

Legend



Stack (top at right): J | I | H | G | F | D | A

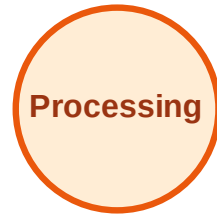
Visited: B, C, E



DFS Traversal

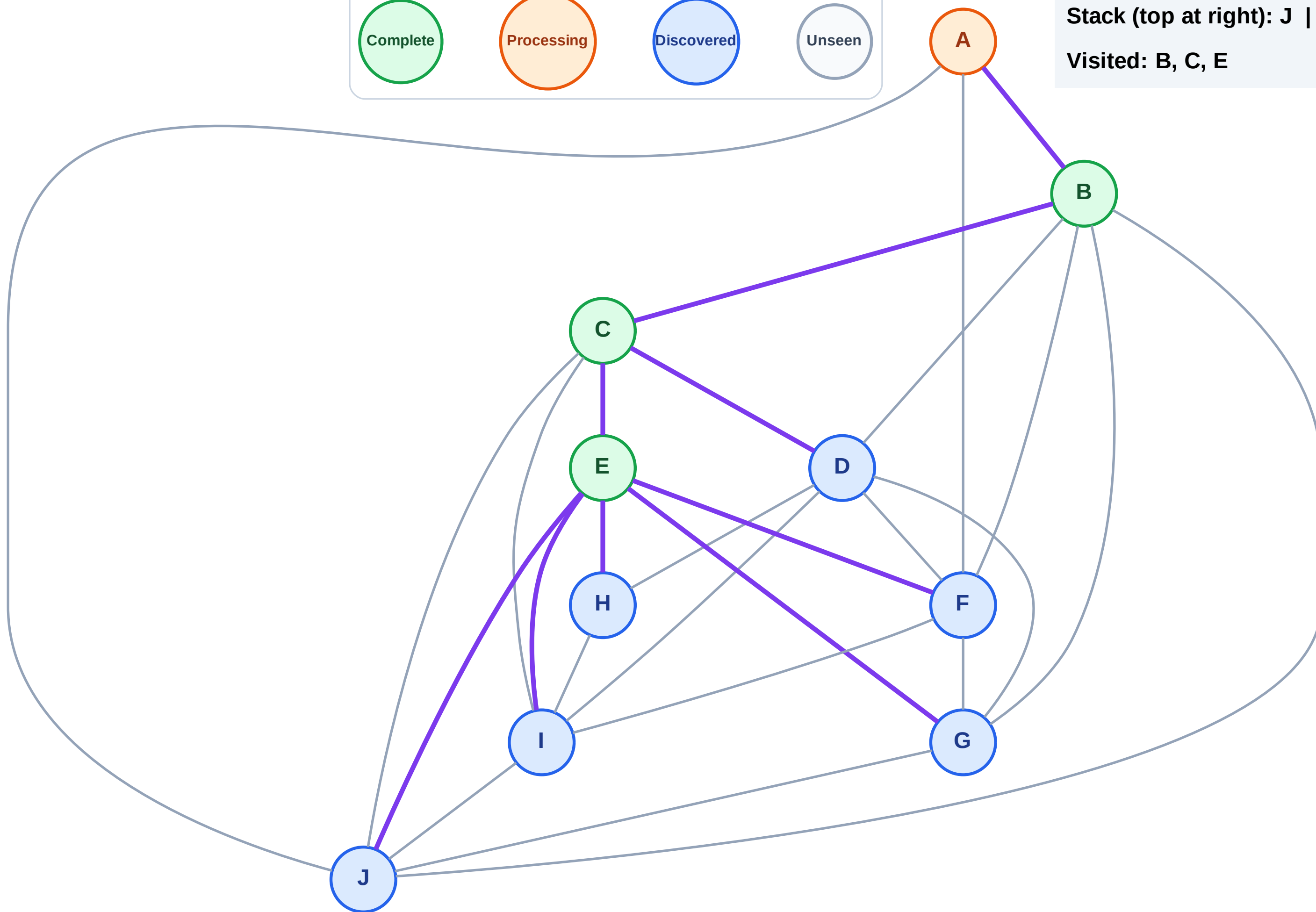
Step 8: Process node A

Legend



Stack (top at right): J | I | H | G | F | D

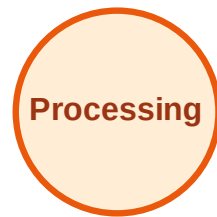
Visited: B, C, E



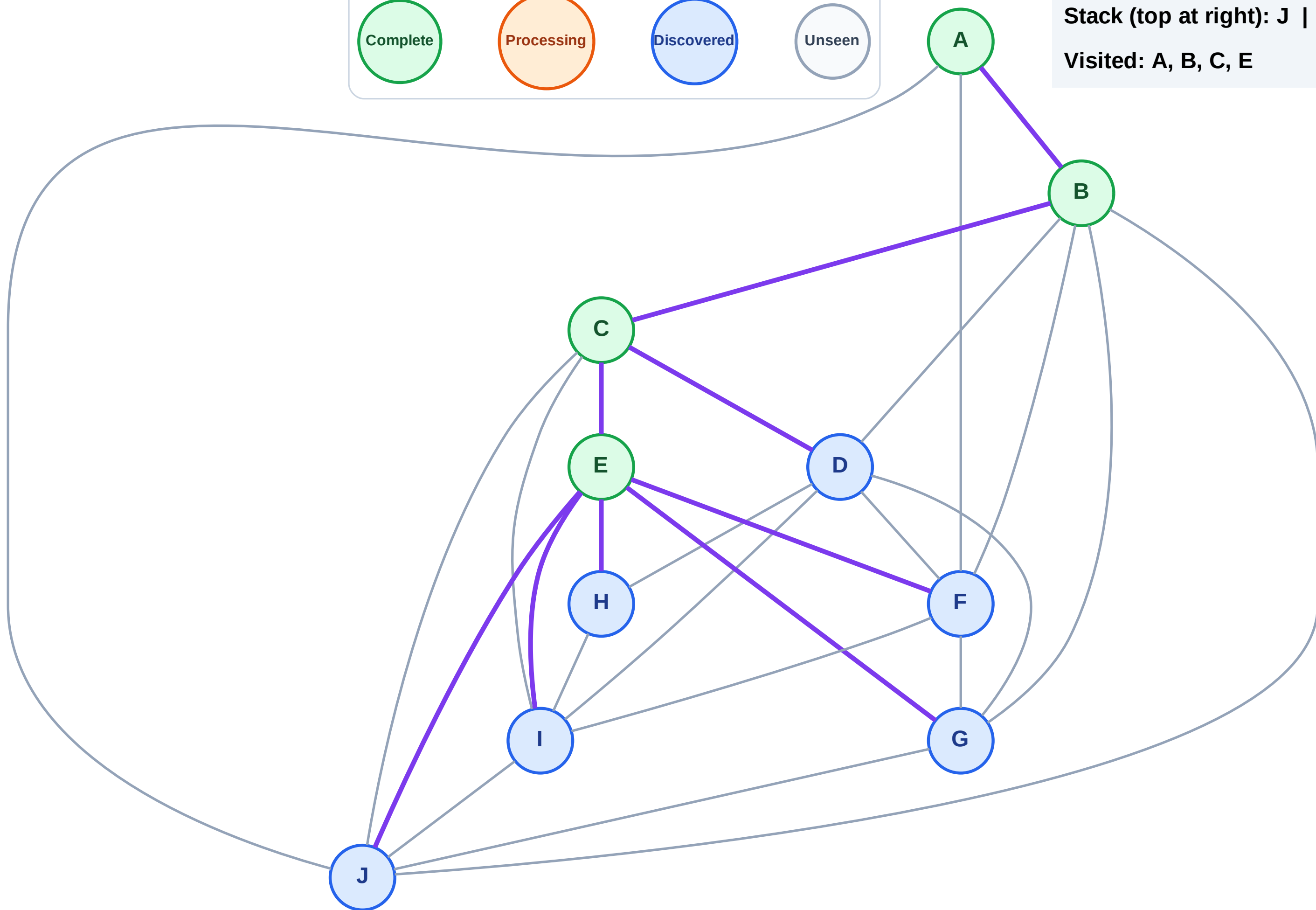
DFS Traversal

Step 9: No new neighbors from A

Legend



Stack (top at right): J | I | H | G | F | D
Visited: A, B, C, E



Step 10: Process node D

and

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle with the label 'Complete', an orange circle with the label 'Processing', a blue circle with the label 'Discovered', and a grey circle with the label 'Unseen'.

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Legend



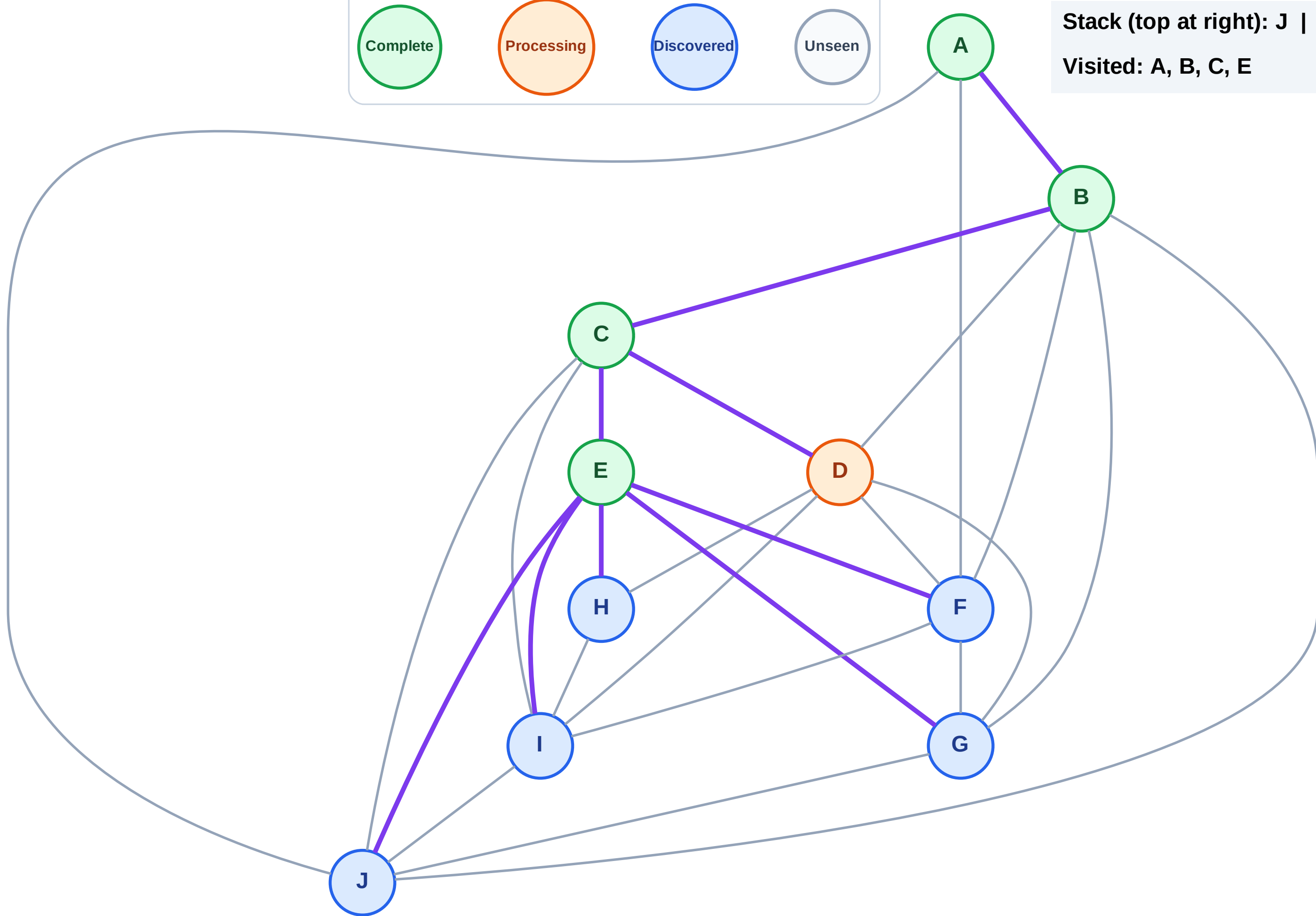
The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Stack (top at right): J | I | H | G | F

Visited: A, B, C, E

Stack (top at right): J | I | H | G | F

Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from D

Legend

Complete

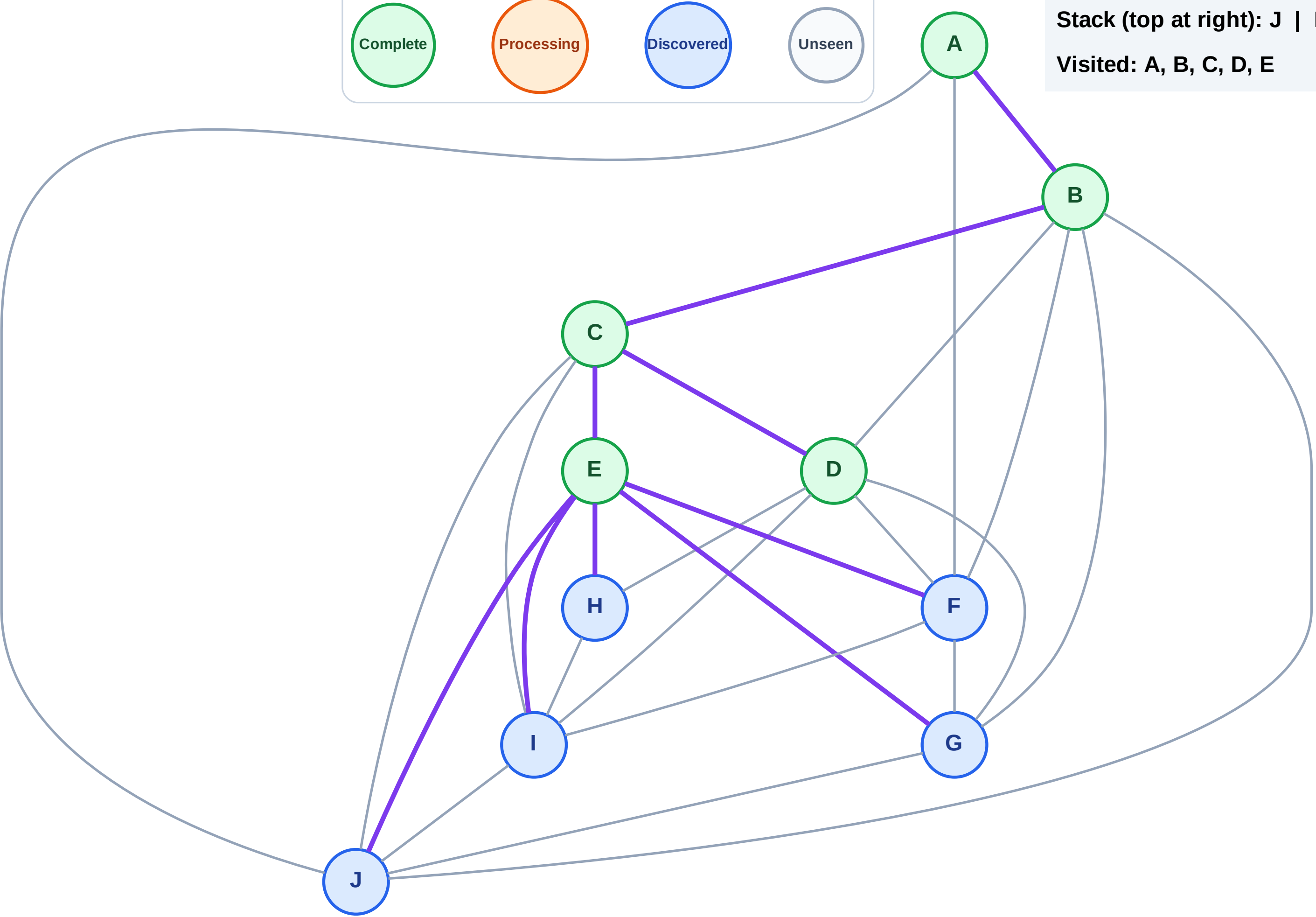
Processing

Discovered

Unseen

Stack (top at right): J | I | H | G | F

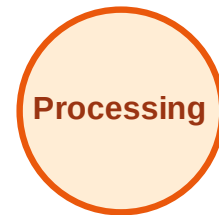
Visited: A, B, C, D, E



DFS Traversal

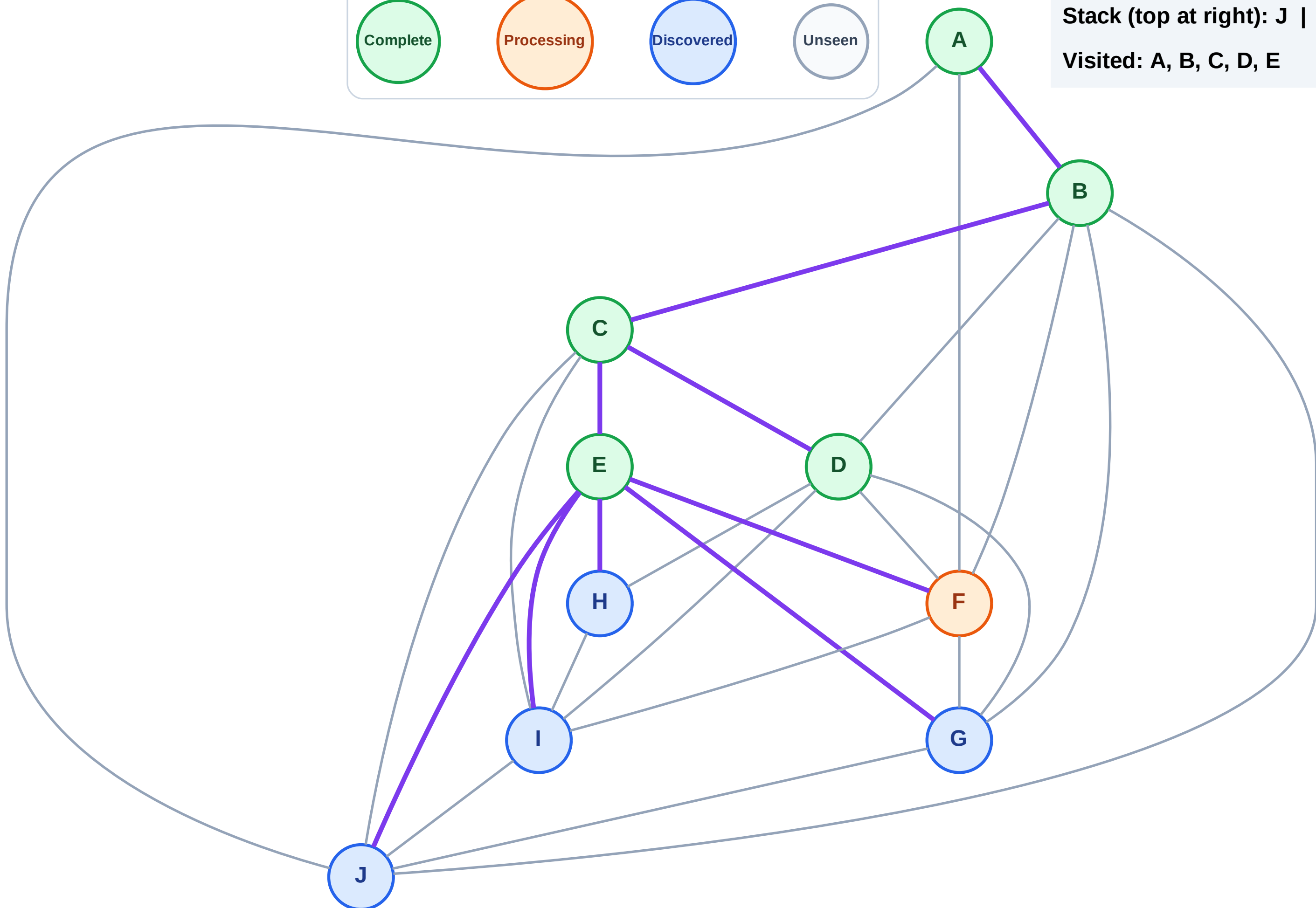
Step 12: Process node F

Legend



Stack (top at right): J | I | H | G

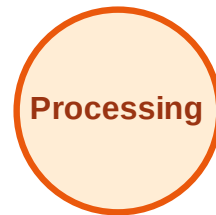
Visited: A, B, C, D, E



DFS Traversal

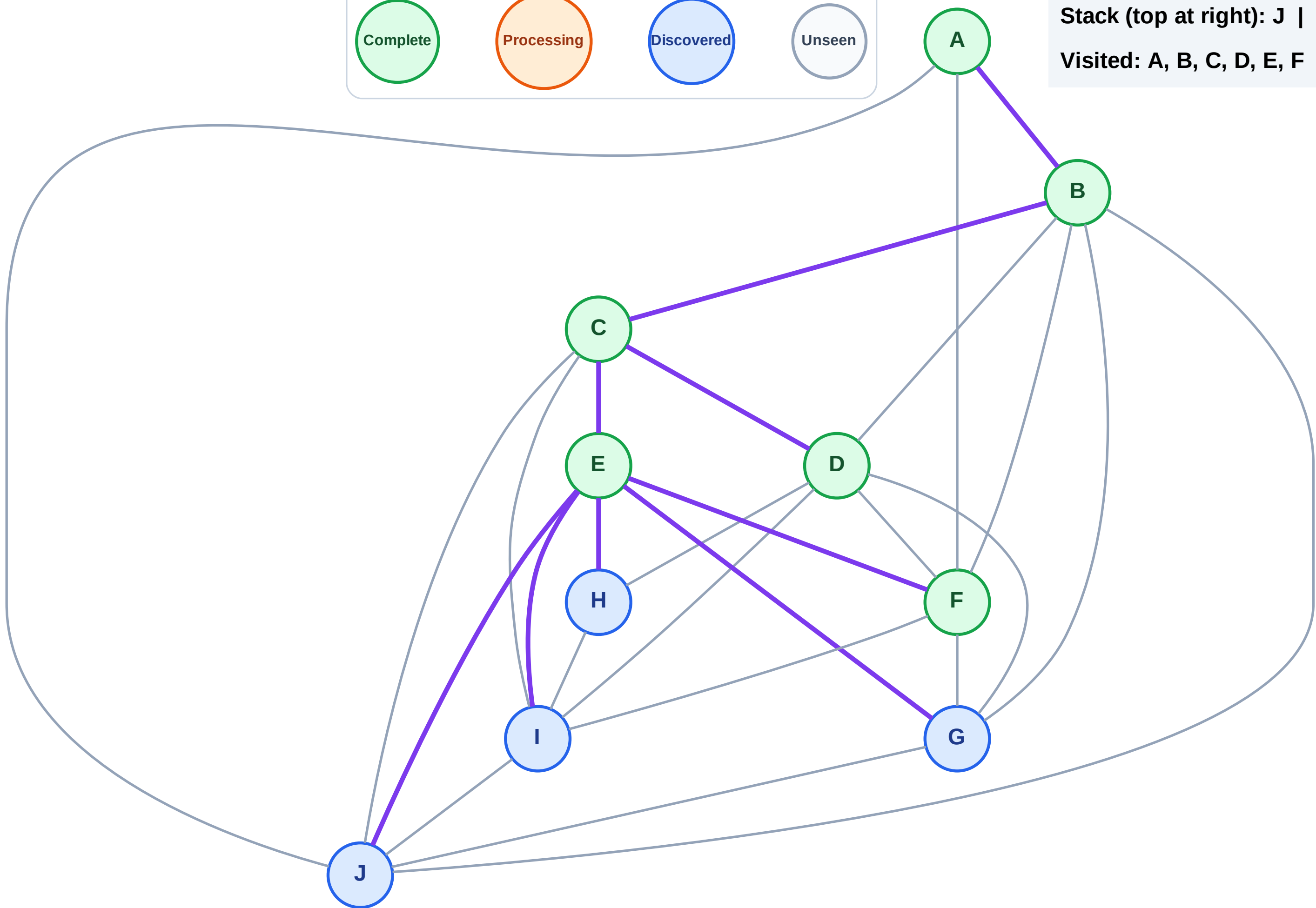
Step 13: No new neighbors from F

Legend



Stack (top at right): J | I | H | G

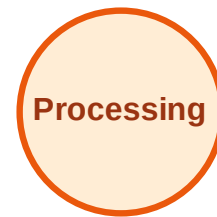
Visited: A, B, C, D, E, F



DFS Traversal

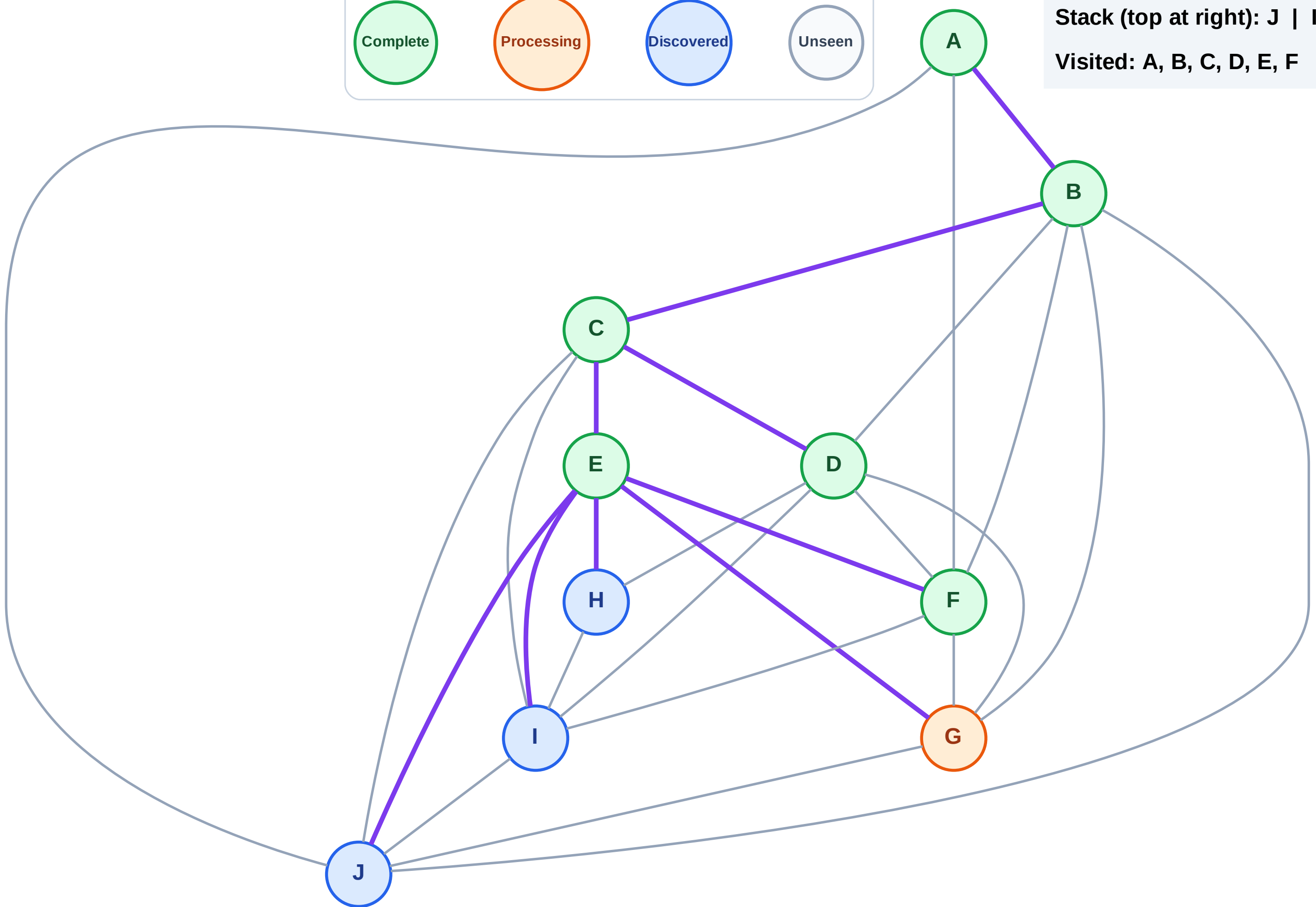
Step 14: Process node G

Legend



Stack (top at right): J | I | H

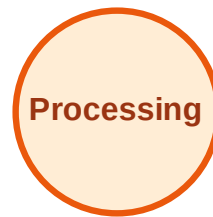
Visited: A, B, C, D, E, F



DFS Traversal

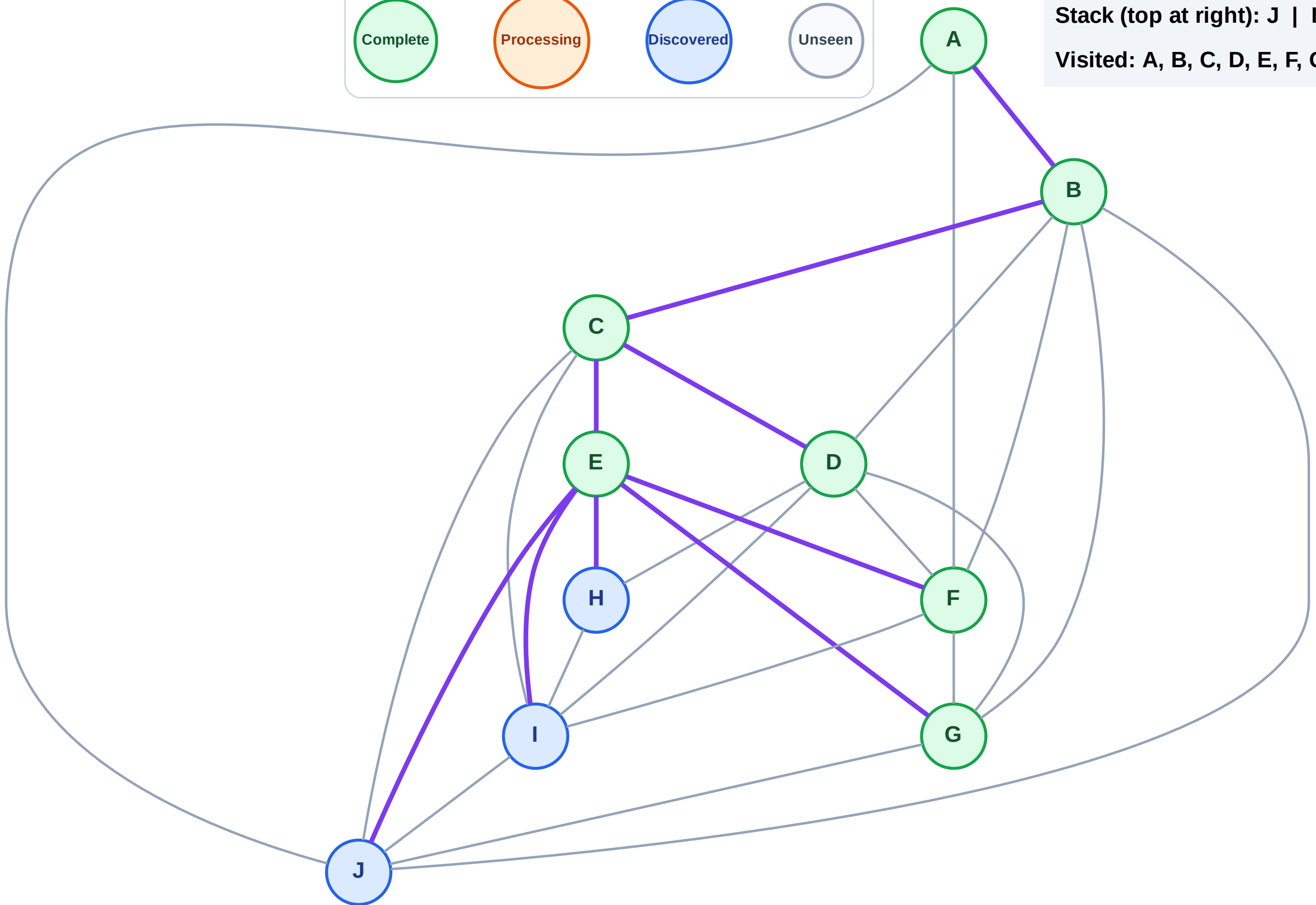
Step 15: No new neighbors from G

Legend



Stack (top at right): J | I | H

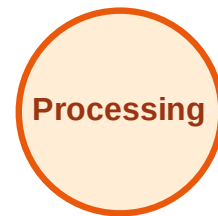
Visited: A, B, C, D, E, F, G



DFS Traversal

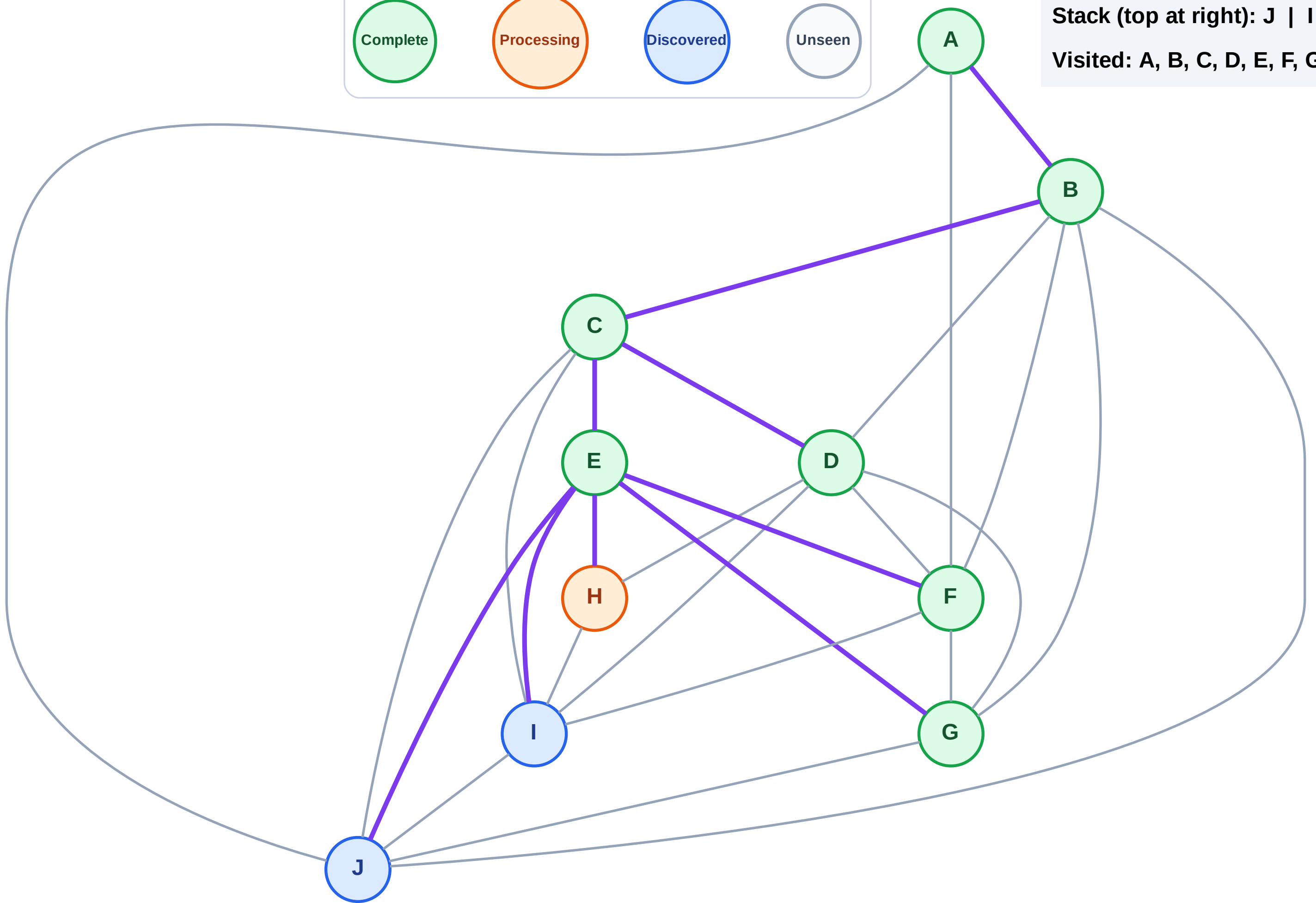
Step 16: Process node H

Legend



Stack (top at right): J | I

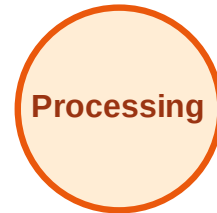
Visited: A, B, C, D, E, F, G



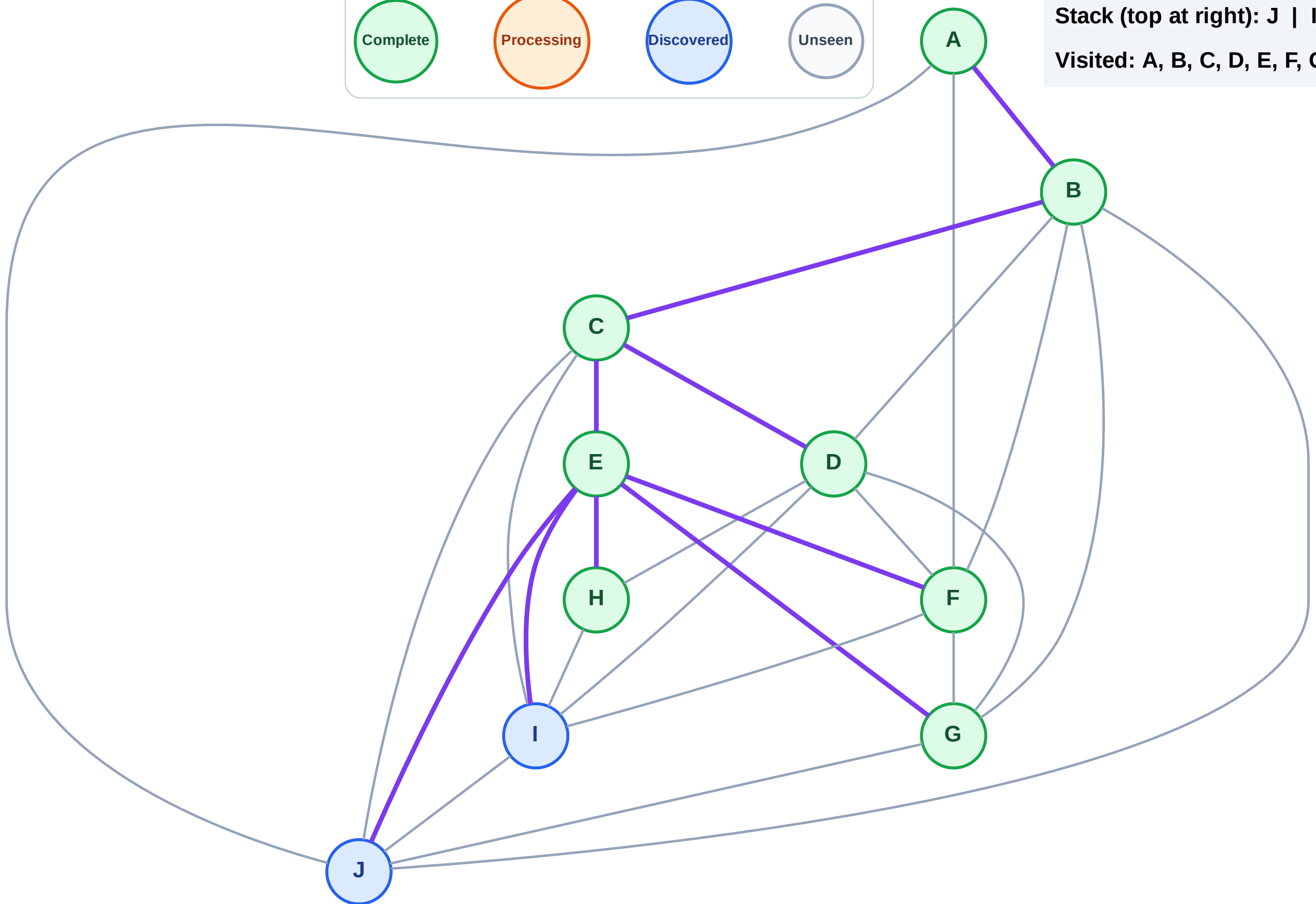
DFS Traversal

Step 17: No new neighbors from H

Legend



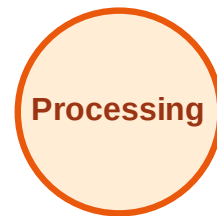
Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

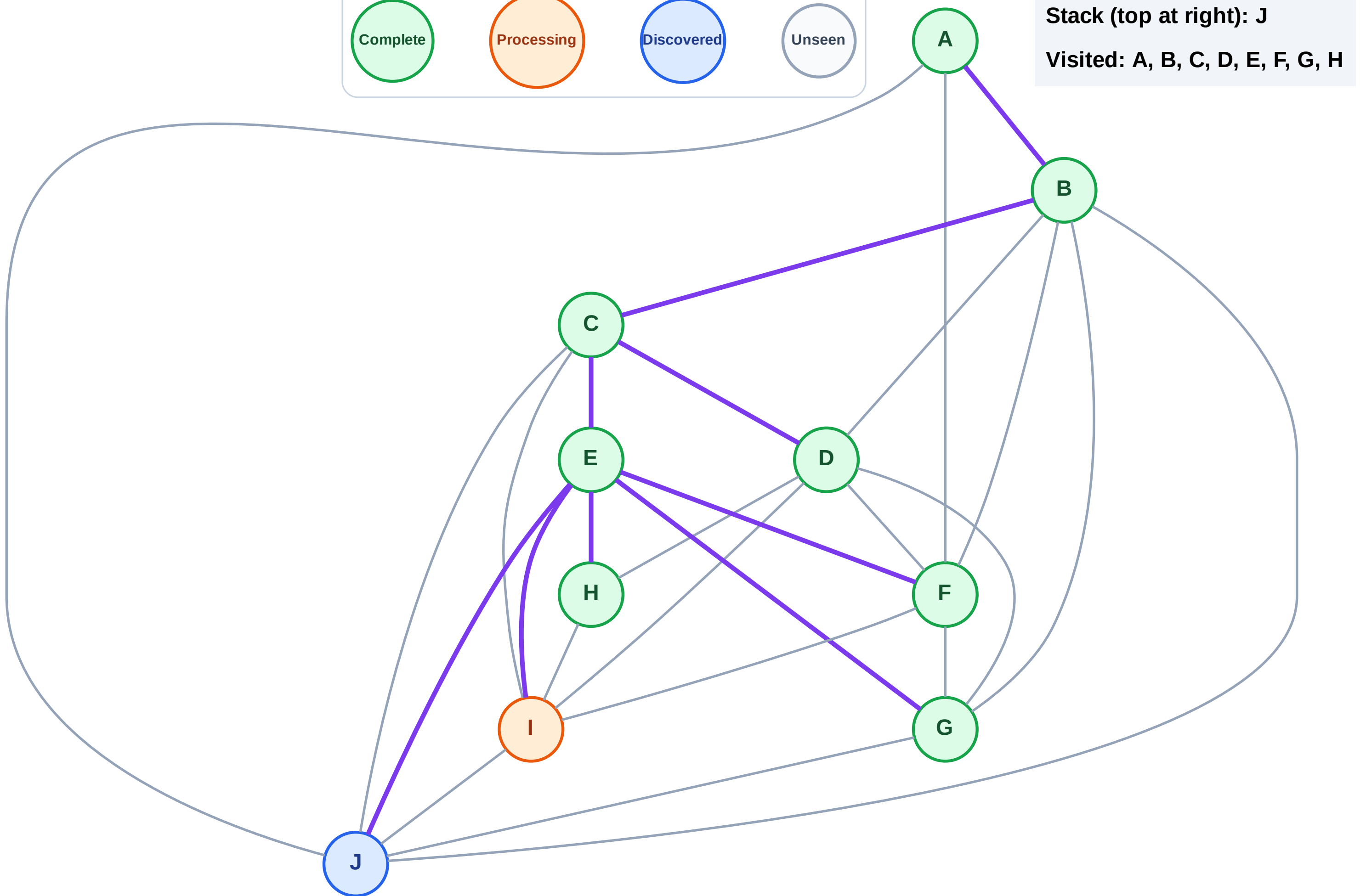
Step 18: Process node I

Legend



Stack (top at right): J

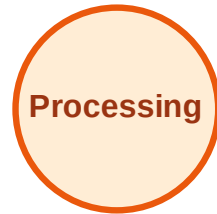
Visited: A, B, C, D, E, F, G, H



DFS Traversal

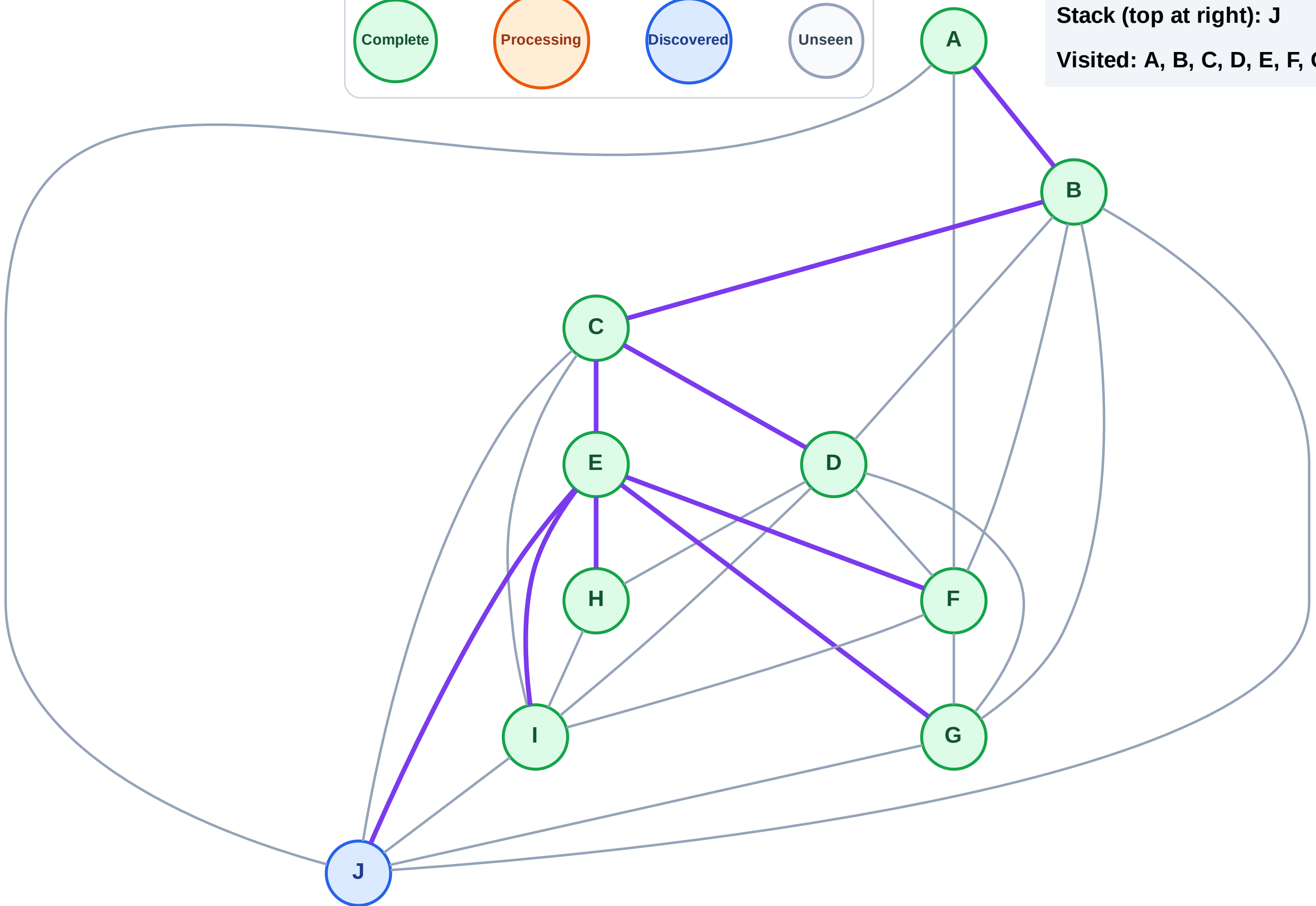
Step 19: No new neighbors from I

Legend



Stack (top at right): J

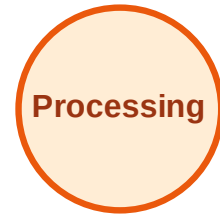
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

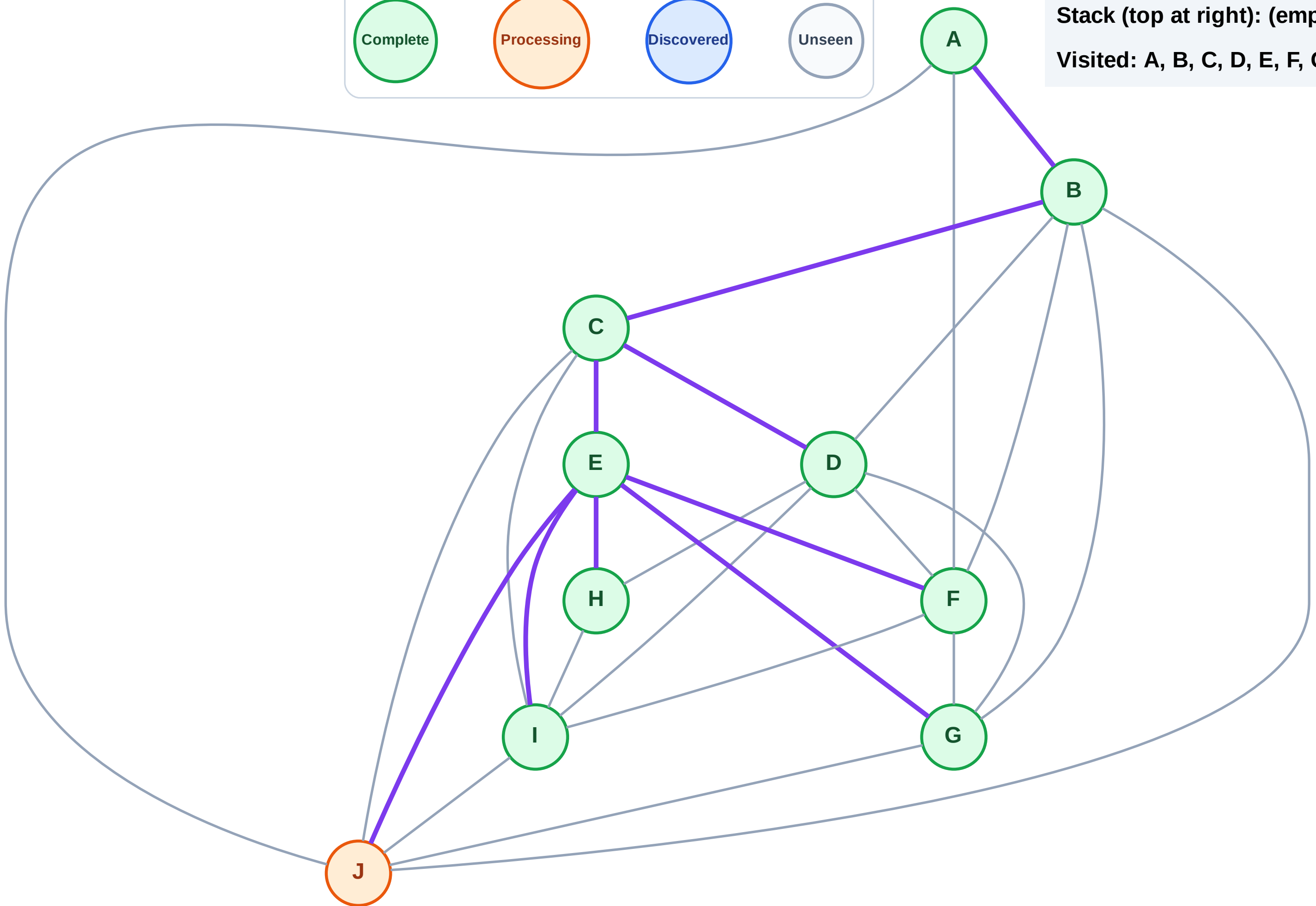
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



Step 21: No new neighbors from J

Step 21: No new neighbors from J

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

